TOPIC 06 · ELECTRICAL PROPERTIES

Electrical Properties

Energy bands and the classification of solids, conduction in metals and its resistivity, non-metallic conductors, intrinsic and extrinsic semiconductors and their applications, and insulating, dielectric, ferroelectric and piezoelectric materials.

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Central idea: the electrical behaviour of a solid is decided by how its electrons fill the energy bands that the atoms form when they come together. A half-filled (or overlapping) band gives a **conductor**; a full valence band separated by a small gap E_g from an empty conduction band gives a **semiconductor**; a large gap gives an **insulator**. In every case the conductivity is carriers times charge times mobility, $\sigma = n\,|q|\,\mu$, and the whole topic is about what controls n (temperature, doping, band gap) and μ (collisions with vibrations, impurities and defects). Insulators that can be polarised (dielectrics, ferroelectrics, piezoelectrics) close the unit.

This page follows the single official session of Topic 6 (Session 13) slide by slide. Slide 1 is the title and agenda (band structure and electrical properties; conducting materials, metallic and non-metallic; semiconducting materials, intrinsic and extrinsic of types n and p; insulating and dielectric materials). Slides 9, 24 and 50 repeat this agenda as section dividers. Every section names the slides it comes from. Figures are redrawn from the slides where a clean redraw is possible, and the charts and tables whose values cannot be redrawn faithfully are copied from the slides as PNG crops; each caption says which. The questions printed on the slides (Q1 to Q5 and Q10) are problems 1 to 5 and 10 of the official sheet, and they are solved on the problems page of this topic.

§ 1 Learning objectives and classification of materials [Session 13 Slides 2, 5]

The slides set three objectives [Session 13 Slide 2]:

- Know the **classification of the materials** according to their electrical and magnetic properties.
- Understand how the **structure and composition** of a material determine its response to an electric field and to a magnetic field.
- Understand the **influence of the temperature** on the electrical and magnetic properties.

Materials are classified by their electrical conductivity σ, measured in $(\Omega \cdot \mathrm{m})^{-1}$ [Session 13 Slide 5]:

Class	Conductivity σ in $(\Omega \cdot \mathrm{m})^{-1}$	Example
Insulators	10^{-20} to 10^{-10}	Polyethylene (PVC coating on the slide)
Semiconductors	10^{-6} to 10^{4}	Si, Ge (microprocessors)
Conductors	about 10^{7}	Metals (copper wire)

Counting from the smallest to the largest value of the table, the scale spans about 27 orders of magnitude (the author's count, not printed on the slide).

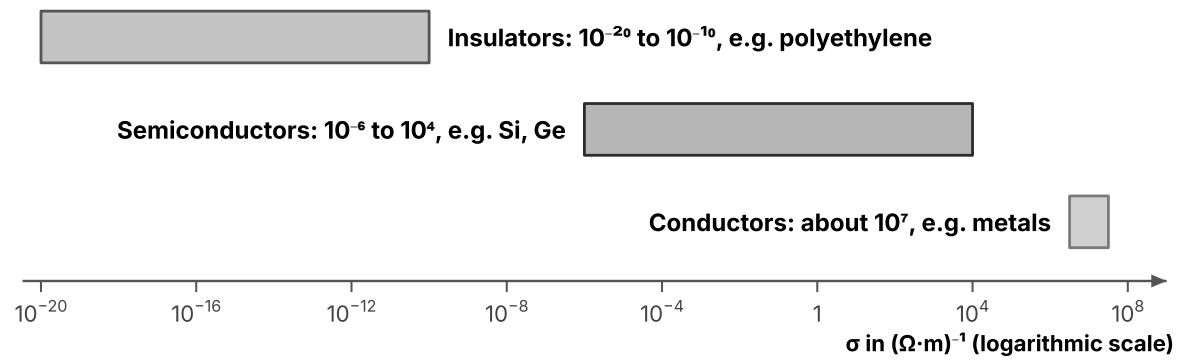


Figure 1. Insulators, semiconductors and conductors on a logarithmic conductivity scale, with the ranges printed on the slide (schematic, not to scale).

The reciprocal of the conductivity is the **resistivity** $\rho = 1/\sigma$, in $\Omega \cdot \mathrm{m}$ [Session 13 Slide 13].

§ 2 Band structures of solids [Session 13 Slides 3-4]

2.1 From atomic levels to bands [Session 13 Slide 3]

An isolated atom has discrete energy levels. When atoms come within close proximity, their orbitals interact and **each atomic level splits**: two atoms give two levels, a few atoms give a few close levels, and a large number of atoms (a solid) gives so many levels, so close in energy, that they form a **continuous energy band**. The slide plots the electron energy against the interatomic separation for an aggregate of atoms, with the core-level electrons staying almost unchanged and the outer levels broadening into bands. Between bands a **band gap** E_g may remain, where no electron states exist.

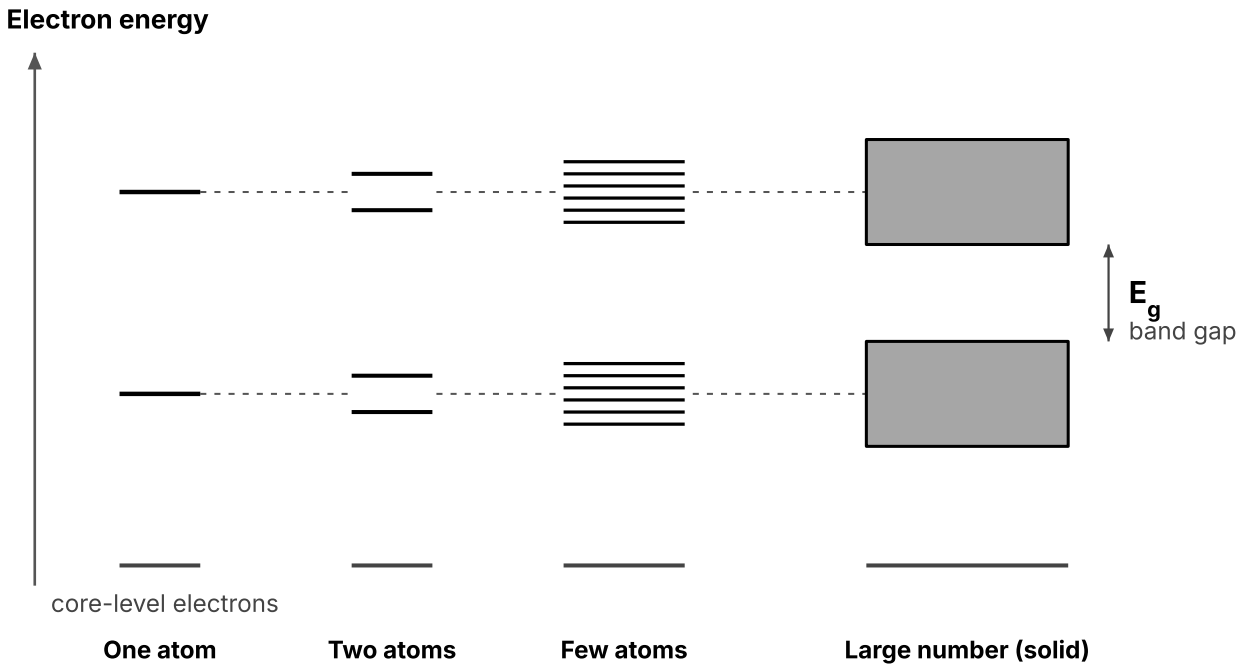


Figure 2. Atomic energy levels split as atoms approach each other: two atoms give two levels, many atoms give a quasi-continuous band, and a gap Eg remains between bands (schematic, not to scale).

2.2 Valence band, conduction band and gap [Session 13 Slide 4]

The electron energy band structure of a solid is given by its **interatomic separation**: the bands broaden as the atoms approach and the solid sits at the **equilibrium interatomic distance**, where the bands have the width and the gap shown in the figure. Two bands matter:

- **Valence band:** the outermost band occupied with electrons (the valence electrons).
- **Conduction band:** the band located immediately above the valence band.
- **Band gap** E_g: the energy range between them where there are no states.

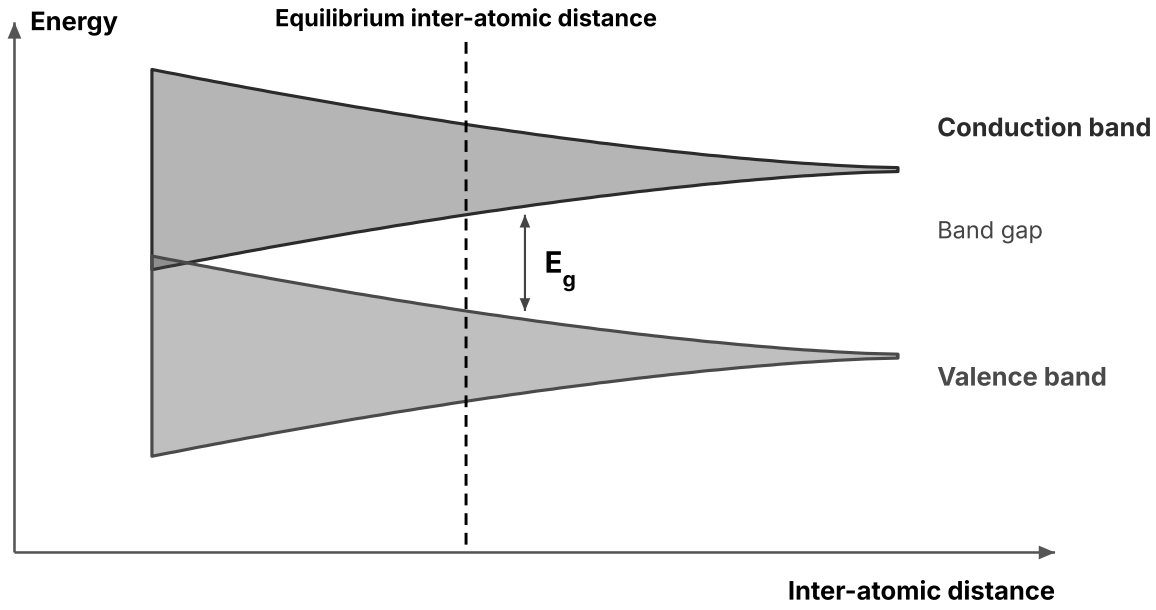


Figure 3. The bands widen as the inter-atomic distance decreases; at the equilibrium distance the valence and conduction bands are separated by the band gap Eg (schematic, not to scale).

§ 3 Band structure of conductors, semiconductors and insulators [Session 13 Slides 5-8]

3.1 Conductors [Session 13 Slides 6-7]

A material conducts when electrons can find **empty states just above the filled ones**, so that a small excitation (an electric field) lets them move. There are two ways [Session 13 Slide 6]:

- **A1) Half-filled valence band.** There is only one s electron per atom (Na, Cu, Al in the slide's caption), so the valence band is half filled: filled states below the Fermi energy E_f and empty states above it in the same band.
- **A2) Overlapping bands.** In Mg the filled $3s$ band overlaps the empty $3p$ band, so the two bands together are only partly filled.

The slide lists the electron configurations used to decide this:

Element	Atomic number	Configuration
Mg	12	$1s^2\,2s^2\,2p^6\,3s^2$
Al	13	$1s^2\,2s^2\,2p^6\,3s^2\,3p^1$
Cu	29	$1s^2\,2s^2\,2p^6\,3s^2\,3p^6\,3d^{10}\,4s^1$

PRINTED CONFIGURATION OF ALUMINIUM

The slide's table gives aluminium as $3s^2\,3p^1$ (one electron in the $3p$ level), while its caption says that Na, Cu and Al have a half-filled band "since there is only one s electron". The table is kept as printed; Na is not in the table.

For a metal, the occupancy of the electron states changes by an electron excitation as in the figure: before the excitation the states are filled up to E_f, and after it an electron has been promoted to an empty state just above E_f [Session 13 Slide 7].

3.2 Semiconductors and insulators [Session 13 Slide 8]

- In a **semiconductor** the filled valence band is separated from the empty conduction band by a relatively **narrow** gap, $E_g < 2$ eV (pure silicon: $E_g = 1.1$ eV; pure germanium: $E_g = 0.67$ eV).
- In an **insulator** the gap is relatively **large**, $E_g > 2$ eV (pure diamond: $E_g = 6$ to 7 eV).

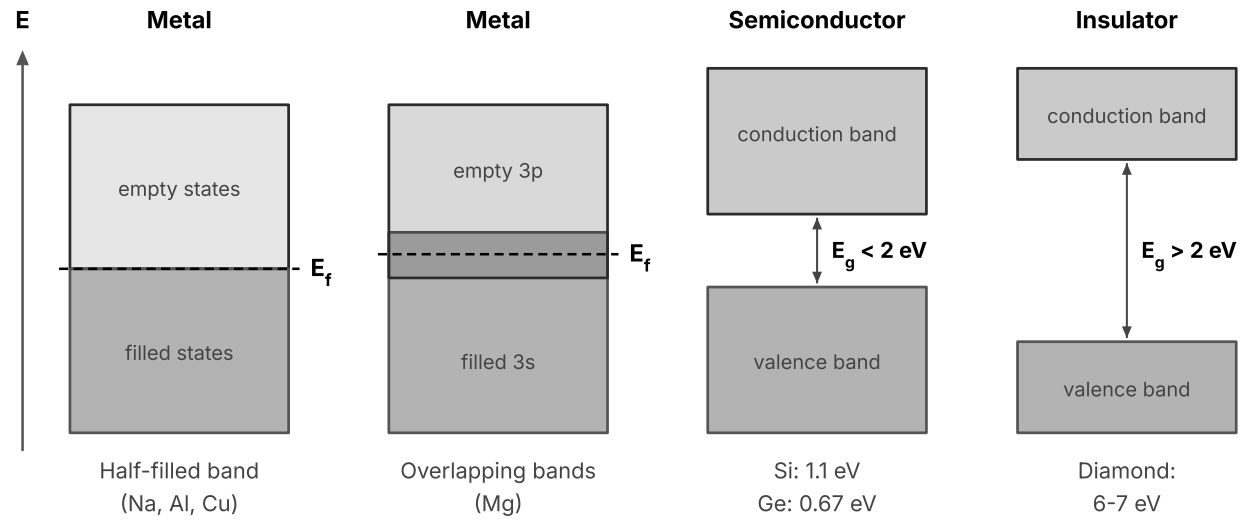


Figure 4. Band structure of conductors (half-filled band, overlapping bands), semiconductors (narrow gap) and insulators (wide gap) (schematic, not to scale).

§ 4 Metallic conductors: Ohm's law and the free-electron model [Session 13 Slides 10-13]

4.1 Electrical and ionic conduction [Session 13 Slide 10]

The macroscopic law is Ohm's law, $I = V/R$, with I the electric current (A), V the potential difference (V) and R the resistance (Ω). The charge can be carried in two ways: by **ionic conductivity** (aqueous electrolytes, some ceramics) or by **electrical (electronic) conductivity** (metals). In a conductive metal there is a cloud of electrons around the nuclei (the positive ions).

For a bar of length l and cross-section A the microscopic form is $\mathbf{J} = \sigma \mathbf{E}$. Writing $E = V/l$ and $J = I/A$ in it gives $V = (l/\sigma A)\, I$, so $R = \rho\, l/A$ (derived here: the slide only writes $I = V/R$; this is the link with the geometry of the specimens of the sheet).

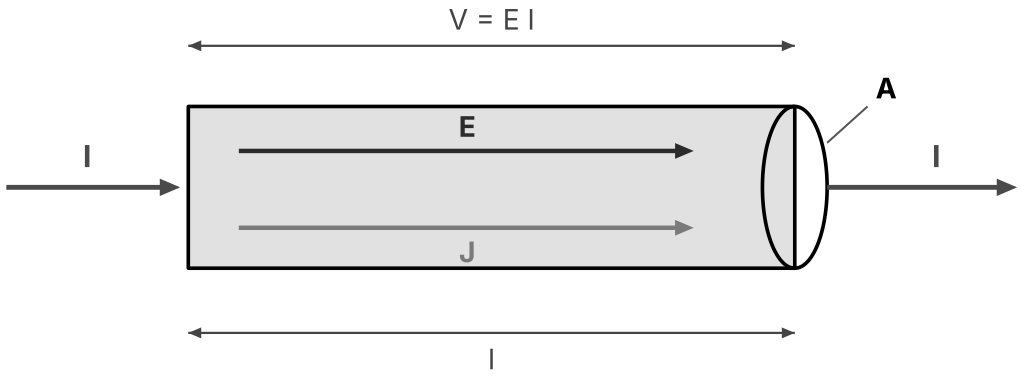


Figure 5. A bar of length l and cross-section A carries a current I under a voltage V: E = V/l, J = I/A, so the resistance is R = ρ l / A (schematic).

4.2 Acceleration of the electrons [Session 13 Slide 10]

When a uniform electric field E is applied, the electrons accelerate. Newton's second law for one electron of charge q_e (1.6×10^{-19} C) and mass m_e (9.109×10^{-31} kg) is

$$F = q_e E = m_e a \quad \Rightarrow \quad a = \frac{q_e E}{m_e}$$

PRINTED UNITS

The slides write the field E in V/m² (and in A/m² on Slide 12). The field strength is in **V/m** (volt per metre); the units printed are typographical slips and V/m is used here.

4.3 Collisions, drift velocity and mobility [Session 13 Slide 11]

If the electrons only accelerated, the current would grow without limit. For a given E the current I is constant, so the speed of the electrons is constant: there are "friction forces", due to **collisions** with the ions, that counteract the acceleration. This is the origin of the **resistivity**. Between collisions the drift velocity grows linearly with time; at each collision it falls back (sawtooth of the figure), so the **average drift velocity** v_d is the average speed of the electrons in the direction of the force generated by the field:

$$v_d = \frac{\tau\,|q_e|\,E}{m_e} = \mu\,E \qquad \text{with} \qquad \mu = \frac{\tau\,|q_e|}{m_e}$$

τ is the **relaxation time** (the mean time between collisions) and μ is the **electron mobility**, in m²/(V·s).

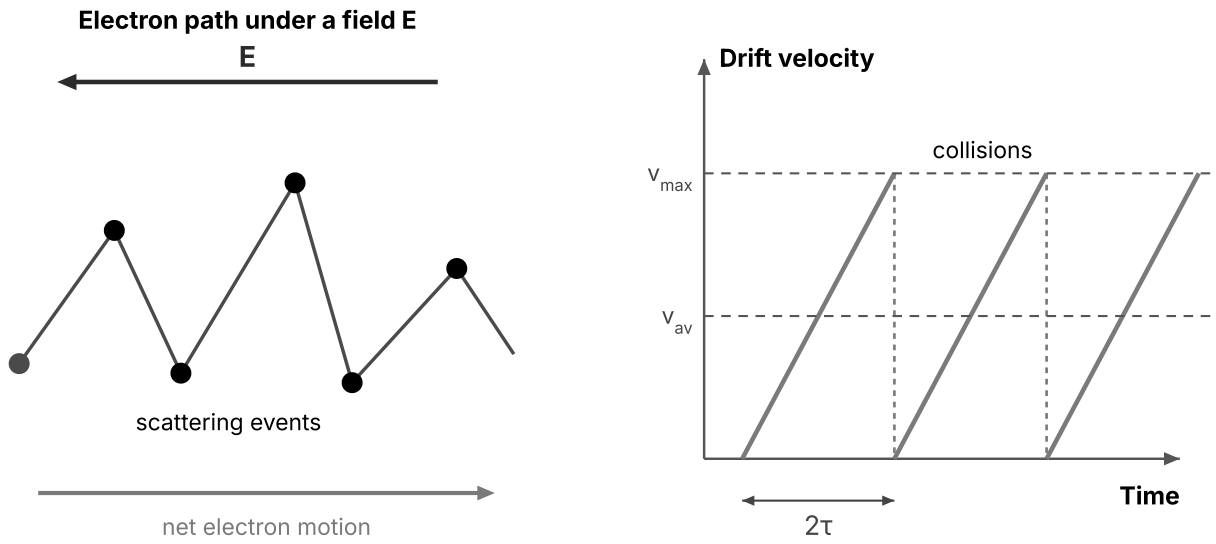


Figure 6. Left: path of a conduction electron with scattering events under a field E. Right: drift velocity against time, which restarts after each collision (classical model, schematic).

4.4 Current density, conductivity and resistivity [Session 13 Slides 12-13]

For a material with n free conducting electrons per unit volume, the current density is the charge that crosses a unit area per second:

$$J = n\,|q_e|\,v_d = n\,|q_e|\,\mu\,E \qquad \text{and, since} \quad J = \sigma E: \qquad \sigma = n\,|q_e|\,\mu$$

In metals the carriers are the non-bonded electrons, so the conductivity depends on **the number of electrons per unit volume and on their mobility**. The resistivity is $\rho = 1/\sigma$, in $\Omega \cdot \mathrm{m}$. The slide adds that the resistivity grows with the **disorder in the metal structure** and with the **number of collisions between the non-bonded electrons and the metallic ions** [Session 13 Slide 13].

§ 5 Electrical resistivity of metals [Session 13 Slides 14-15]

5.1 Matthiessen's rule [Session 13 Slide 14]

The total resistivity of a metal is the sum of a thermal part and a residual part:

$$\rho_{total} = \rho_T + \rho_r$$

ρ_T is the **thermal resistivity**, produced by the vibrations of the lattice, and ρ_r is the **residual resistivity**, produced by impurities and defects (next section). The slide's plot of resistivity against temperature shows ρ_r as a constant floor (the curve tends to it at low temperature) and ρ_T as the temperature component added on top.

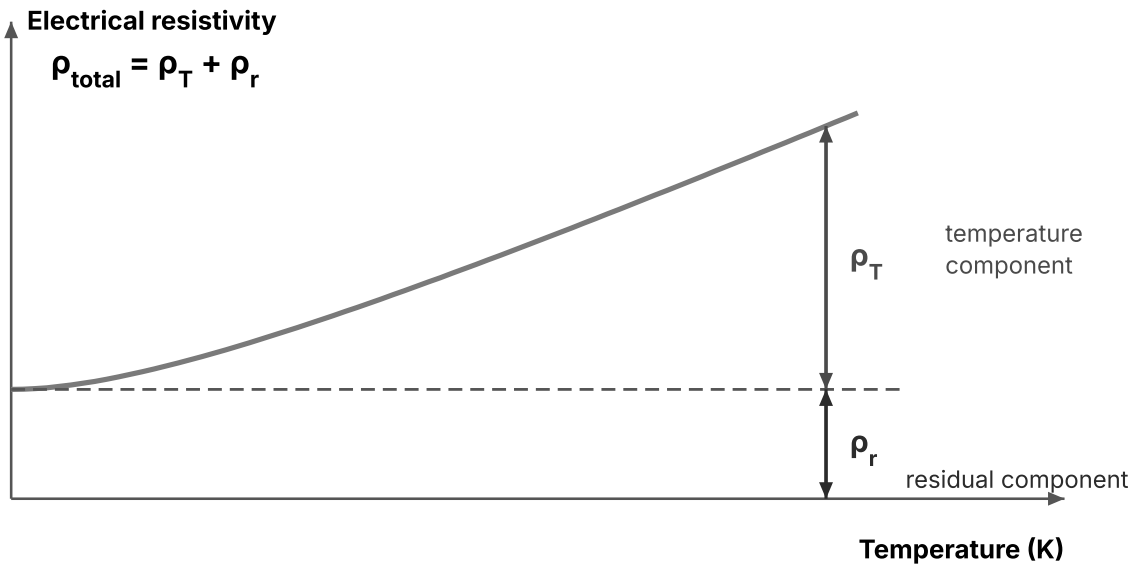


Figure 7. Resistivity of a metal against temperature: a temperature-dependent part ρT on top of a temperature-independent residual part ρr (schematic, not to scale).

5.2 Thermal resistivity [Session 13 Slide 14]

The thermal resistivity grows linearly with the temperature:

$$\rho_T = \rho_0 \left(1 + \alpha_T \, \Delta T\right)$$

ρ_0 is the resistivity at 0 °C (in $\Omega \cdot \mathrm{m}$) and α_T is the temperature coefficient of resistivity (in $^\circ\mathrm{C}^{-1}$). Because ρ_0 is the value at 0 °C, ΔT is the temperature in degrees Celsius. The slide's table:

Metal	Resistivity at 0 °C in $\mu\Omega \cdot \mathrm{cm}$	α_T in $^\circ\mathrm{C}^{-1}$
Aluminium	2.7	0.0039
Copper	1.6	0.0039
Gold	2.3	0.0034
Iron	9	0.0045
Silver	1.47	0.0038

Metal	Electrical resistivity at 0°C ($\mu\Omega \cdot$ cm)	Temperature resistivity coefficient α_T (°C^{-1})
Aluminum	2.7	0.0039
Copper	1.6	0.0039
Gold	2.3	0.0034
Iron	9	0.0045
Silver	1.47	0.0038

Figure 8. Resistivity at 0 °C and temperature coefficient of resistivity of aluminium, copper, gold, iron and silver, copied from Session 13 Slide 14.

The second chart of the slide shows the effect of temperature on the resistivity of selected metals (Fe, Rb, Os, Co, Na, Ir, Rh, Mg, Au, Cu, Ag): almost straight lines above about –100 °C.

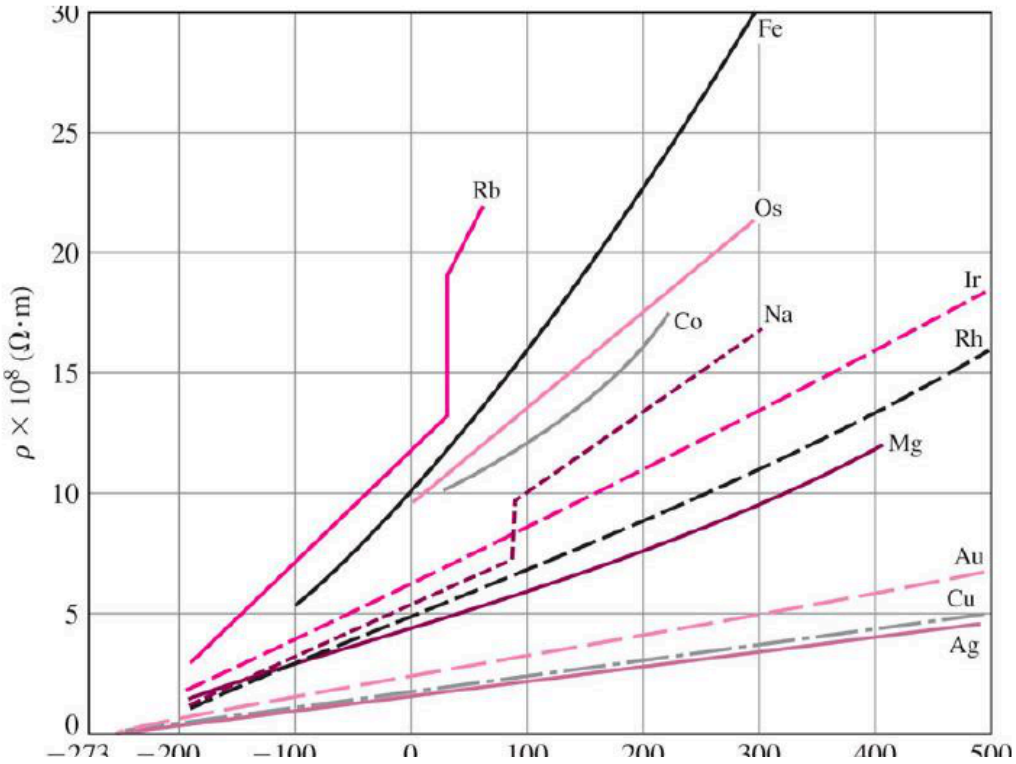


Figure 9. Effect of temperature on the electrical resistivity of selected metals, copied from Session 13 Slide 14.

5.3 Question 1: equal resistivity of silver and aluminium [Session 13 Slide 15]

Q1 asks at which temperature a silver wire has the same resistivity as an aluminium wire at 45 °C, with $\rho_0 = 1.47 \times 10^{-6}$ and $2.70 \times 10^{-6}\ \Omega \cdot \mathrm{cm}$ and $\alpha_T = 0.0038$ and $0.0039\,^{\circ}\mathrm{C}^{-1}$ for silver and aluminium. The slide first evaluates the aluminium resistivity with the law above, $\rho_{Al,45} = 2.7 \times 10^{-6}(1 + 0.0039 \cdot 45) = 3.173 \times 10^{-6}\ \Omega \cdot \mathrm{cm}$, and then solves the law for the temperature of silver, $T = (\rho_T - \rho_{0,Ag})/(\rho_{0,Ag}\,\alpha_{T,Ag}) = 304.86\,^{\circ}\mathrm{C}$. This is problem 1 of the sheet, and its page redoes the calculation with the units in SI and the unrounded intermediate value (the key of the sheet is 305 °C).

§ 6 Residual resistivity: impurities and defects [Session 13 Slides 16-17]

The residual resistivity ρ_r does not depend on temperature. It depends on [Session 13 Slide 16]:

- the **concentration and type of impurities or alloying elements** (impurity resistivity ρ_i), and
- the **structural imperfections** (deformation resistivity ρ_d): dislocations, grain boundaries and similar.

So $\rho_r = \rho_i + \rho_d$ and, with Matthiessen's rule, $\rho_{total} = \rho_T + \rho_i + \rho_d$. ρ_r **increases with the degree of distortion of the metal lattice**: a foreign atom, a large one or a small one, disturbs the periodicity around it. The slide's chart shows how small additions of various elements raise the room-temperature resistivity of copper: for each element the resistivity grows linearly with the weight percent of impurity, with a slope that depends on the element (the steepest are P, Fe, Si and As; Ni, Pb and Ag are the gentlest among those plotted).

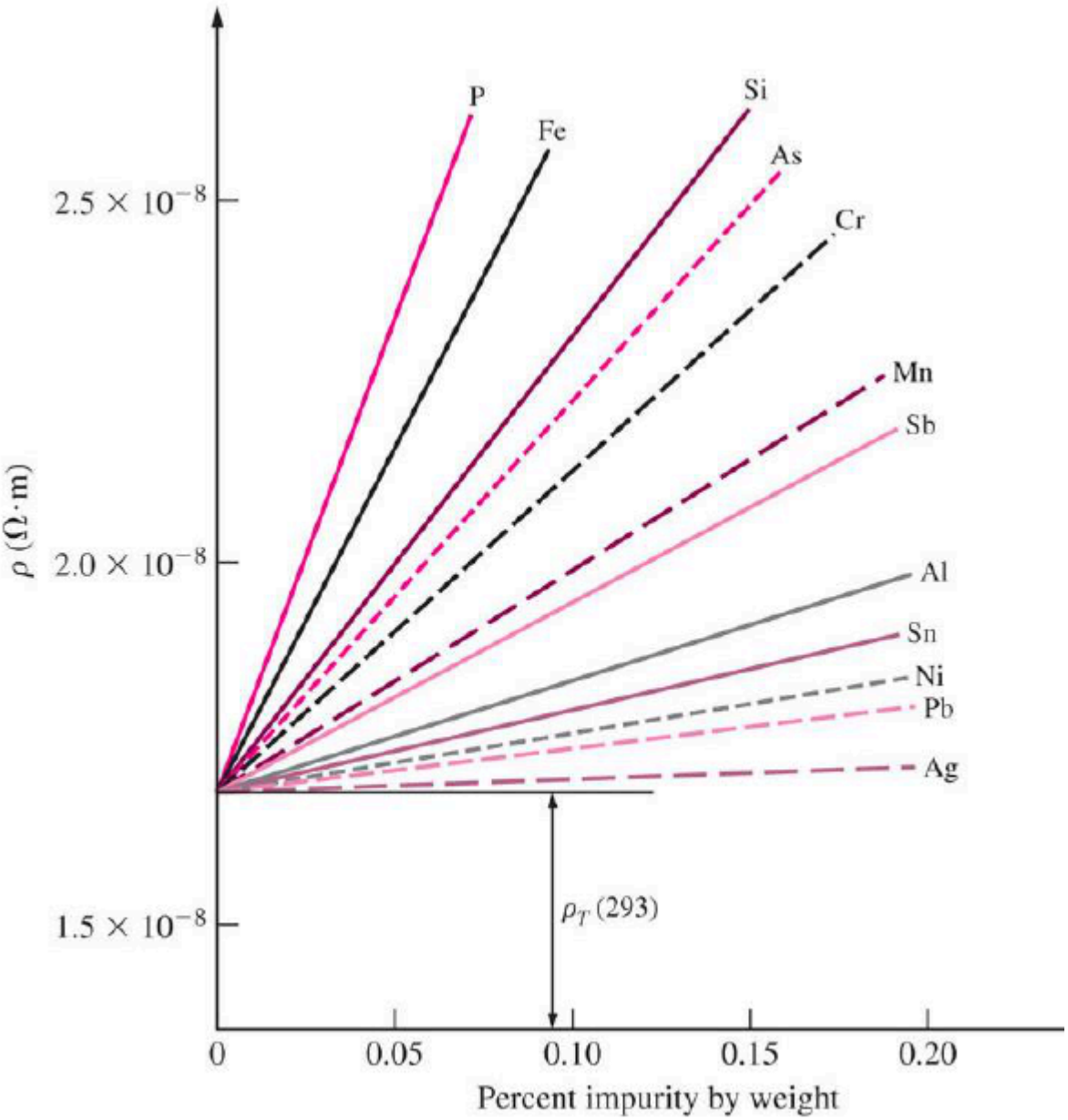


Figure 10. Effect of small additions of various elements on the room-temperature electrical resistivity of copper, copied from Session 13 Slide 16.

6.1 Scattering of the electrons [Session 13 Slide 17]

A conduction electron moves through a **perfect crystal** without scattering. In a crystal **heated to a high temperature** the atoms vibrate and displace from their sites, and in a crystal containing **atomic-level defects** (foreign atoms, vacancies, dislocations) the periodicity is broken too. In both cases the electron is scattered, which reduces its mobility and therefore the conductivity. The chart of the slide shows copper with different nickel contents and deformed copper: the curves are parallel, separated by the constant residual terms ρ_i and ρ_d, while the temperature component ρ_T is the same.

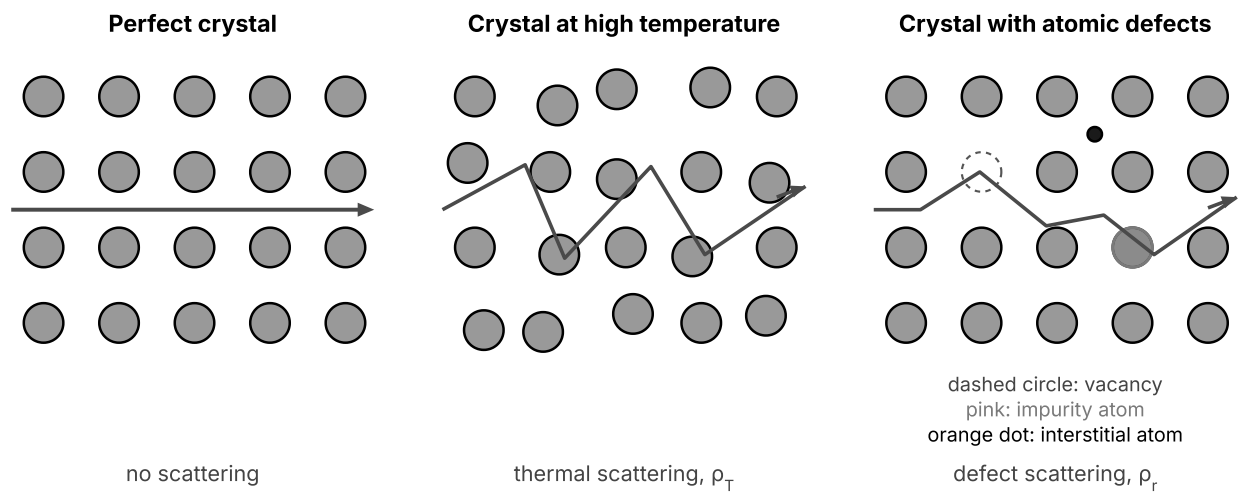


Figure 11. Electron path through a perfect crystal, a crystal at high temperature (vibrating atoms) and a crystal with atomic-scale defects; scattering reduces the mobility and the conductivity (schematic).

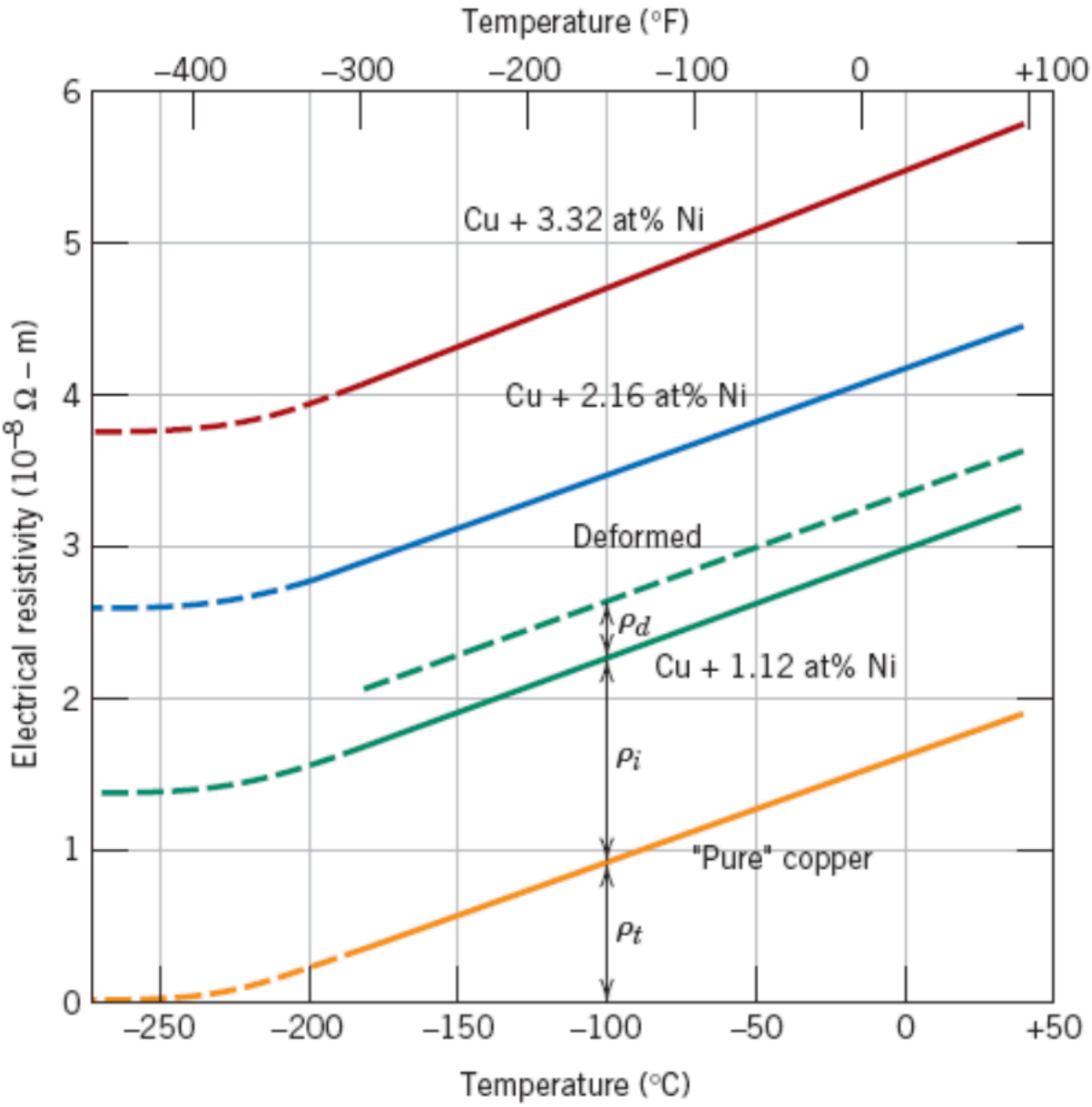
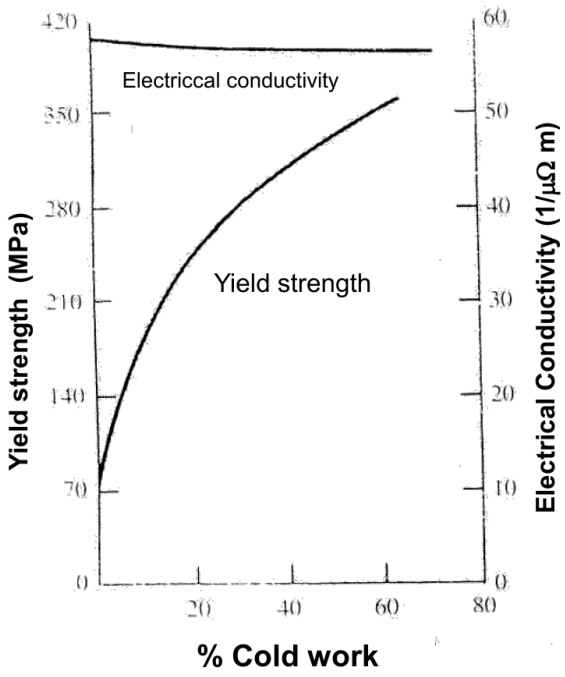


Figure 12. Resistivity against temperature of pure, alloyed (Cu + Ni) and deformed copper, showing the thermal, impurity and deformation parts, copied from Session 13 Slide 17.

§ 7 Mechanical properties and resistivity [Session 13 Slides 18-19]

All hardening methods cause, to a greater or lesser extent, a decrease of the conductivity, [Session 13 Slide 18]. The reason, in the author's reading (not printed on the slide), is that every hardening method of Topic 5 works by putting obstacles in the way of the dislocations, and the same obstacles scatter the electrons. The slide plots yield strength and conductivity against the percent of cold work and against the percent of zinc (a solid solution):

- **Cold work:** the yield strength rises strongly, while the conductivity falls only slightly (the chart compares with the zinc case, where it falls much more).
- **Zinc in solution:** the yield strength rises moderately but the conductivity falls sharply.



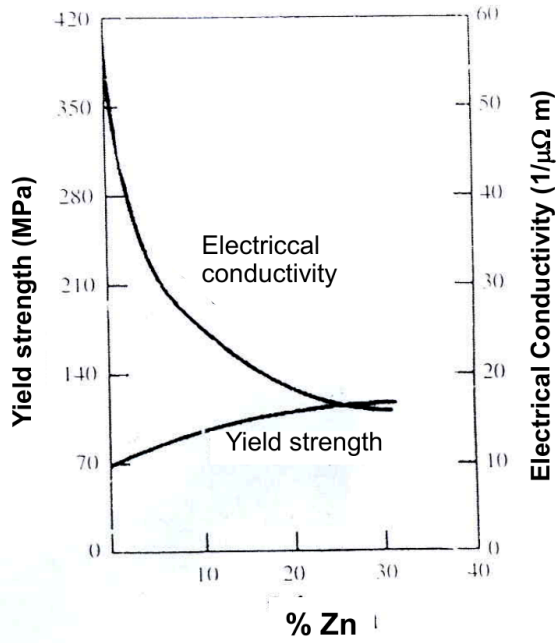


Figure 13. Yield strength and electrical conductivity against percent cold work (left) and percent Zn (right), copied from Session 13 Slide 18.

Q2 on Slide 19 asks what happens to the conductivity and the tensile strength of an aluminium alloy wire when it is alloyed with copper, when its working temperature is lowered, when its grain size is increased, when intermetallic precipitates are formed and when it is cold worked, and which strategy gives the lightest wire with the same strength and conductivity. It is problem 2 of the sheet.

§ 8 Non-metallic conductors [Session 13 Slides 20-23]

8.1 Conductive polymers [Session 13 Slide 20]

A few thermoset polymers, when they are specifically manufactured, can become conductive. Their advantages against metals are **flexibility and easy formability**. The main mechanism for transmitting the charge is the movement of **delocalised electrons**: an exchange reaction of electrons between neighbouring redox sites.

8.2 Graphene [Session 13 Slide 21]

Graphene is a single layer of a hexagonal lattice formed by hexagons with an atom at each vertex (graphite is a stack of these layers). It has a huge π cloud with delocalised electrons that can move all over the surface at a speed higher than in any other material (about 1000 km/s). The slide compares

Material	**Conductivity in** $(\Omega \cdot \mathrm{m})^{-1}$
Graphene	0.96×10^{8}
Cu	0.60×10^{8}
Si	4.5×10^{-4}

PRINTED SYMBOL

The slide writes these values with the symbol ρ, but the unit $(\Omega \cdot \mathrm{m})^{-1}$ is the unit of conductivity, so they are σ.

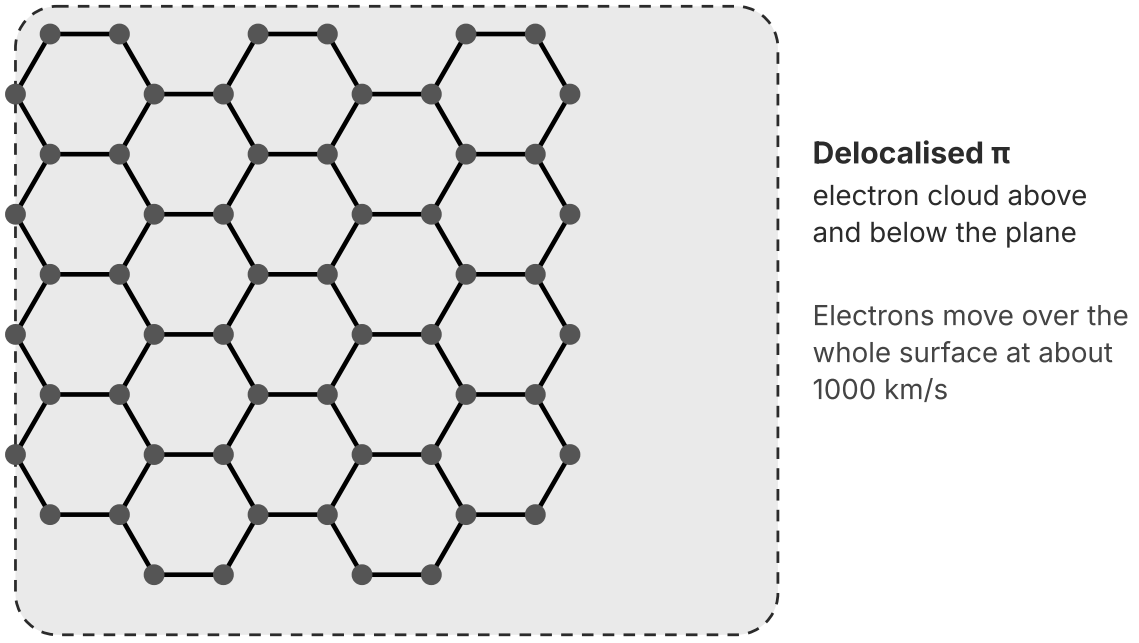


Figure 14. Graphene: a single layer of carbon atoms on a hexagonal lattice, with a π cloud of delocalised electrons (schematic).

8.3 Conductive ceramics [Session 13 Slide 22]

Ceramics are traditionally insulators, but some advanced ceramics are good electrical conductors. The main mechanism for transmitting the charge is

- **Electronic:** movement of electrons due to impurities with a different valence, or rare-earth elements. Uses: resistors, electrodes or heating elements.
- **Ionic:** movement of anions or cations through vacancies or interstitial voids at high temperatures. Uses: gas sensors, fuel cells or batteries. The slide shows two pictures: the movement of ions or, equivalently, the movement of vacancies, and the movement of small interstitial cations ($r_i < r_{atom}$), and a solid oxide fuel cell (anode, electrolyte with oxygen ions, cathode).

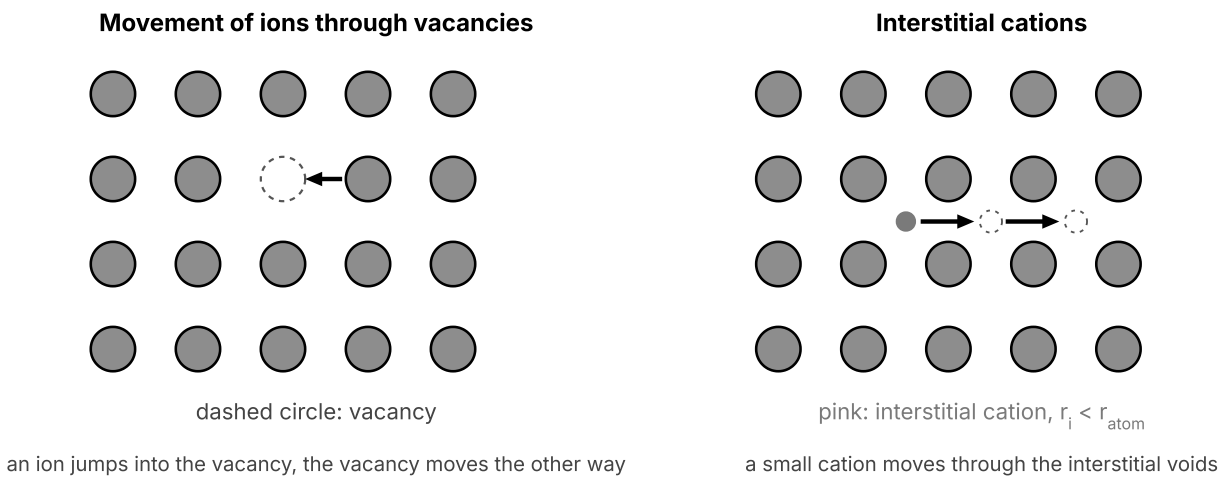


Figure 15. Ionic conduction: an ion hops into a neighbouring vacancy (left), or a small interstitial cation moves through the interstitial voids (right) (schematic).

Q3 on Slide 23 compares the effect of vacancies, interstitial cations, interstitial anions, temperature, disorder of the lattice and doping with other valences on an ionic ceramic and on a metal. It is problem 3 of the sheet.

§ 9 Semiconductors: definition and types [Session 13 Slides 25-26]

Semiconductors are materials whose electrical conductivity is approximately midway between the values for good conductors and insulators. In the energy-band model the electronic band structure consists of a **full valence band and an empty conduction band, separated by an energy gap** $E_g < 2$ eV [Session 13 Slide 25]. The materials are mediocre conductors, but their conductivity is **easily controllable**, and this became useful in the twentieth century with the manufacture of materials of extremely high chemical and structural purity.

Two types depending on the origin of the carriers [Session 13 Slide 26]:

- **Intrinsic semiconductors:** the conductivity is determined by the inherent properties of the pure material (elemental silicon and germanium).
- **Extrinsic semiconductors:** the electrical properties are given by adding **dopants** with a number of valence electrons different from the solvent. They are generally substitutional solid solutions with low impurity concentrations (100 to 1000 ppm). The extra charge carriers are of two kinds, **type n** and **type p**.

§ 10 Intrinsic semiconductors [Session 13 Slides 27-30]

10.1 Excitation and the two kinds of carriers [Session 13 Slides 27-28]

In an intrinsic semiconductor an electric or thermal excitation makes an electron pass from the valence band to the conduction band. This generates a **hole** in the valence band, and **both the free electron and the hole** are charge carriers. In the covalent crystal a bonding valence electron cannot move unless it receives enough energy to leave its bonding position; once a critical amount of energy has been supplied, it becomes a free conduction electron and leaves a **positively charged hole** in the lattice.

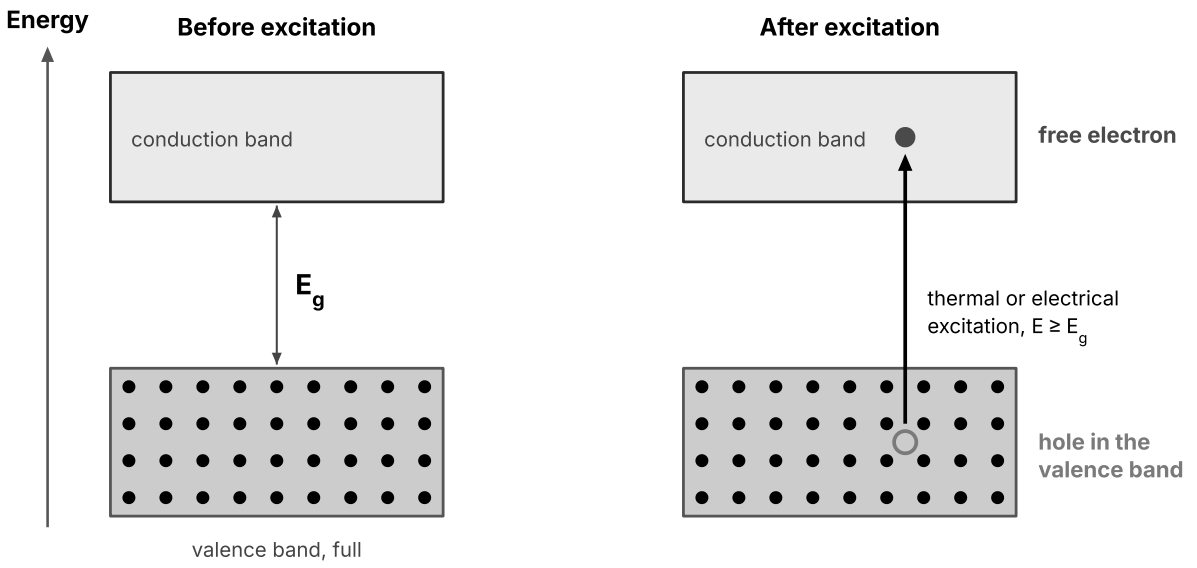


Figure 16. Intrinsic semiconductor: an electron excited across the gap leaves a hole in the valence band, giving two types of charge carrier (schematic, not to scale).

There are two types of carriers: **electrons** in the conduction band (charge -1.6×10^{-19} C, density n_e) and **holes** in the valence band (charge $+1.6 \times 10^{-19}$ C, density n_H). The conductivity is the sum of both contributions:

$$\sigma = n_e \left|q_e\right| \mu_e + n_H \left|q_H\right| \mu_H = n \left|q_e\right| (\mu_e + \mu_H) \qquad \text{since} \quad n_e = n_H = n, \;\; \left|q_e\right| = \left|q_H\right|$$

μ_e and μ_H are the mobilities of electrons and holes, and $|q_e|$ is the absolute value of the electron or hole charge. In an intrinsic semiconductor $\mu_e > \mu_H$ [Session 13 Slide 33].

10.2 Effect of temperature [Session 13 Slide 29]

For an intrinsic semiconductor **both n and p increase with temperature**:

$$n_i \propto \exp\left(-\frac{E_g}{2kT}\right)$$

k is Boltzmann's constant (the sheet gives $k_B = 8.62 \times 10^{-5}$ eV/K), T the temperature in kelvin and E_g the band gap; $n_i = n = p$ is the number of excited electrons (and holes) per unit volume. Since $\sigma \propto n_i$,

$$\sigma = \sigma_0 \exp\left(-\frac{E_g}{2kT}\right) \qquad \Rightarrow \qquad \ln \sigma = \ln \sigma_0 - \frac{E_g}{2kT}$$

Plotted as $\ln \sigma$ against $1/T$ the law is a straight line of slope $-E_g/2k$, as the slide's chart shows. Temperature also disorders the crystal and lowers the mobility, **but this effect is masked by the increase in the number of carriers** [Session 13 Slide 29]. Compare with a metal, where n is fixed and only the mobility changes, so that σ falls with temperature.

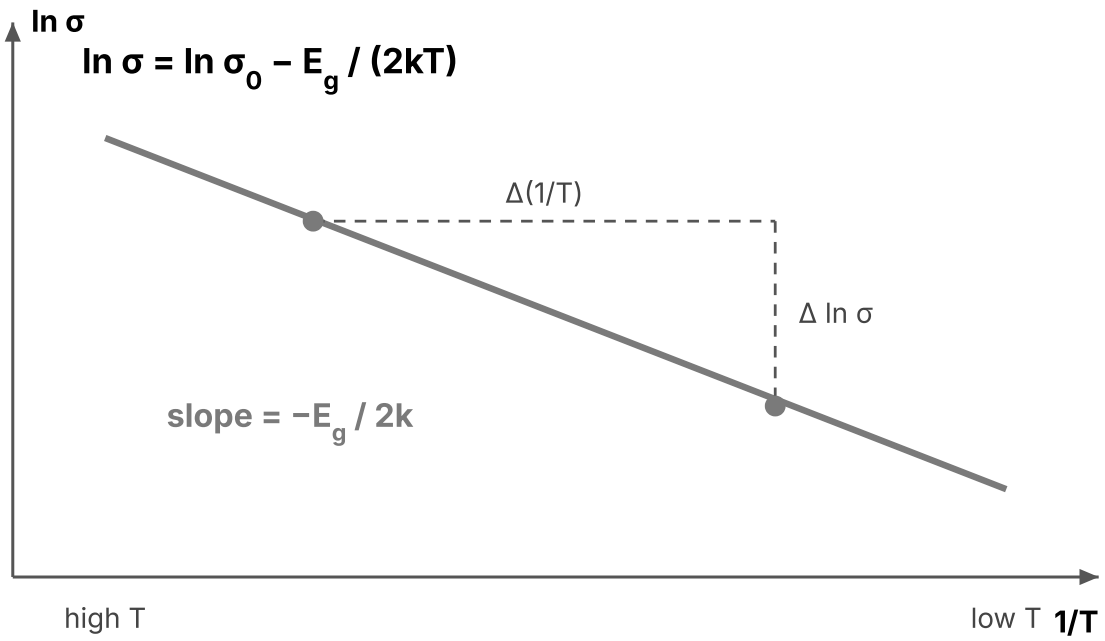


Figure 17. ln σ against 1/T is a straight line of slope −Eg/2k for an intrinsic semiconductor (schematic, not to scale).

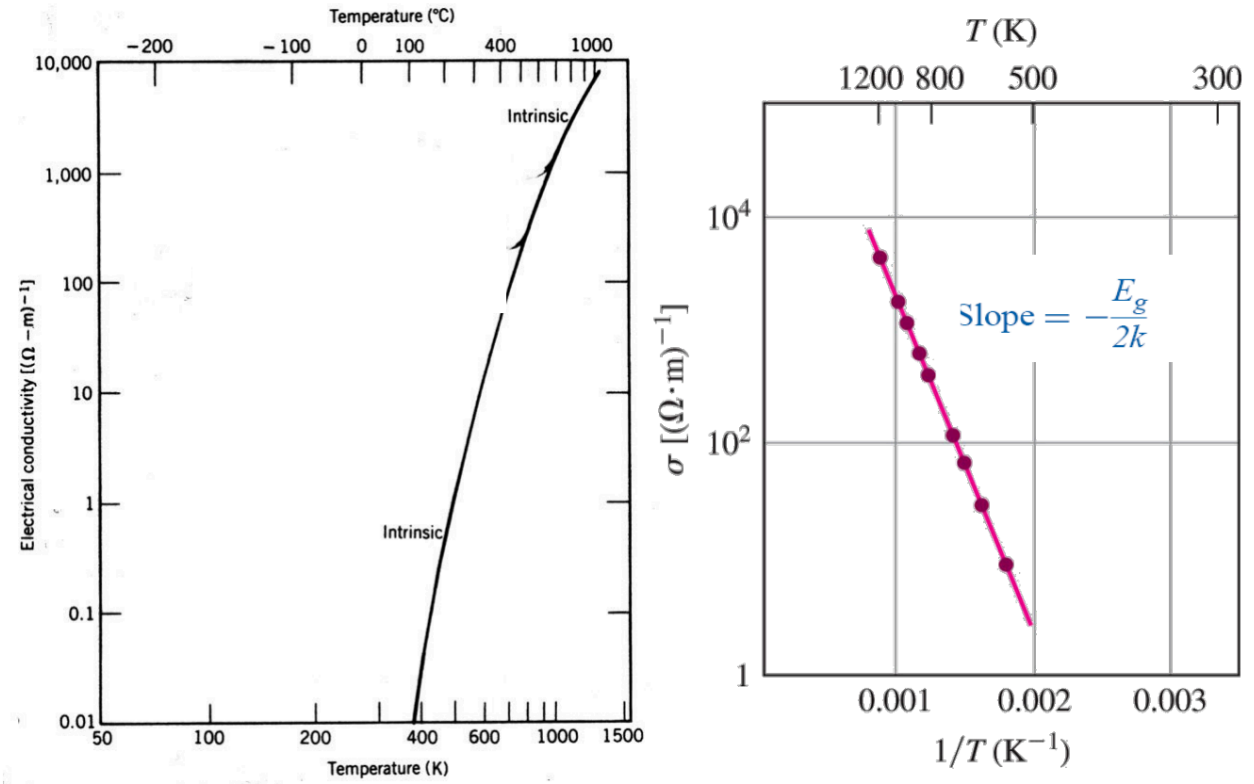


Figure 18. Electrical conductivity against temperature (left) and against 1/T with the slope −Eg/2k (right) of an intrinsic semiconductor, copied from Session 13 Slide 29.

Q4 on Slide 30 asks to justify the different effect of a temperature increase on a metal and on an intrinsic semiconductor. It is problem 4 of the sheet, and problems 7 (germanium at 75 °C) and 8 (carrier concentrations of silicon) apply the formulas of this section.

§ 11 Intrinsic semiconductors: elemental and compound materials [Session 13 Slides 31–34]

11.1 Elemental semiconductors [Session 13 Slide 31]

The elemental semiconductors are the pure elements of **group IVA**, silicon and germanium, bonded by **covalent bonds**. As the atom size increases, E_g decreases, and the conductivity of the intrinsic semiconductor increases. The figure of the slide shows the bands of Si and Ge at 0 K (no electrons in the conduction band, gap printed as 1.09 eV for Si and 0.72 eV for Ge) and at 300 K, when a few electrons have crossed the gap. The data table of the slide:

Property	Silicon	Germanium
Energy gap, eV	1.1	0.67
Electron mobility μ_n, m²/(V·s)	0.135	0.39
Hole mobility μ_p, m²/(V·s)	0.048	0.19
Intrinsic carrier density n_i, carriers/m³	1.5×10^{16}	2.4×10^{19}
Intrinsic resistivity ρ, Ω·m	2300	0.46

Property	Silicon	Germanium
Density, g/m³	2.33×10^6	5.32×10^6

	Silicon	Germanium
Energy gap, eV	1.1	0.67
Electron mobility μ_n, $m^2/(V \cdot s)$	0.135	0.39
Hole mobility μ_p, $m^2/(V \cdot s)$	0.048	0.19
Intrinsic carrier density n_i, carriers/m^3	1.5×10^{16}	2.4×10^{19}
Intrinsic resistivity ρ, $\Omega \cdot m$	2300	0.46
Density, g/m^3	2.33×10^6	5.32×10^6

Figure 19. Energy gap, mobilities, intrinsic carrier density, intrinsic resistivity and density of Si and Ge, copied from Session 13 Slide 31.

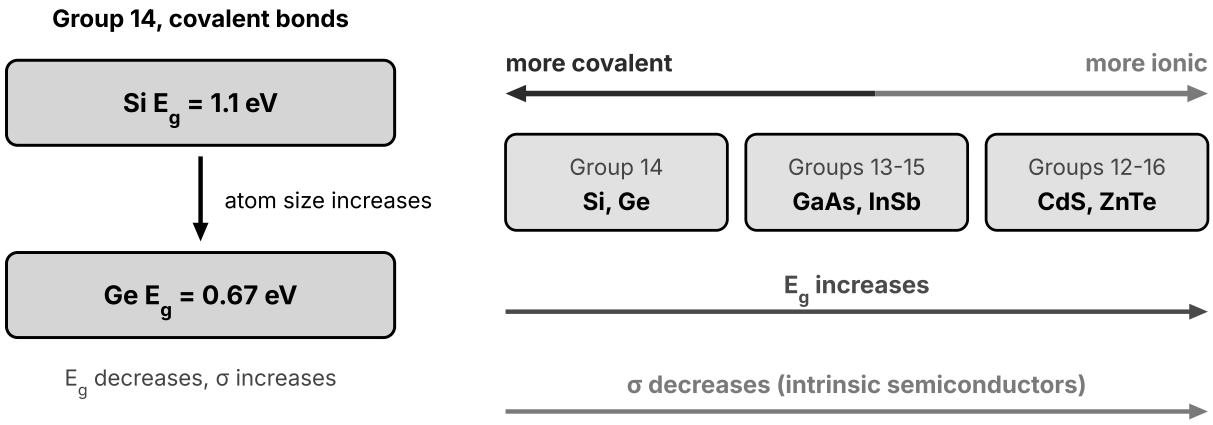


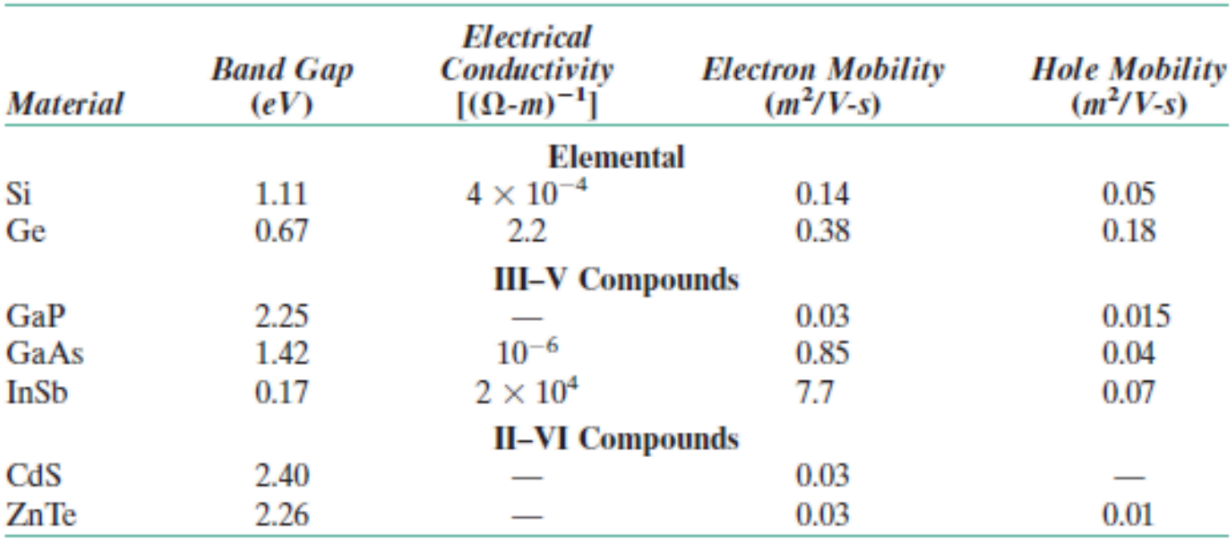
Figure 20. Trends of the band gap: it falls as the atom size grows in group 14 and rises as the bonding becomes less covalent and more ionic; the conductivity follows the opposite trend (schematic).

11.2 Compound semiconductors [Session 13 Slides 32-33]

Compounds of elements of groups IIIA and VA (GaAs, InSb) and of groups IIB and VIA (CdS, ZnTe) are also intrinsic semiconductors. Going from the group-14 elemental semiconductors (purely covalent) to the 13-15 and then to the 12-16 compounds, **the covalent character decreases and the ionic character increases**: E_g **increases** and, for an intrinsic semiconductor, σ **decreases**. The slide also classifies semiconductors as simple (Si, Ge) and compound (III-V: GaAs, InP, GaN; II-VI: ZnS, ZnSe; IV-IV: SiC, SiGe).

The conductivity of an intrinsic semiconductor is determined by n: the number of charge carriers is small, but the high chemical and structural purity favours high mobilities. The table of the slide (it prints slightly different Si and Ge mobilities from the previous one):

Material	E_g in eV	σ in $(\Omega \cdot \mathrm{m})^{-1}$	μ_e in m²/(V·s)	μ_H in m²/(V·s)
Si	1.11	4×10^{-4}	0.14	0.05
Ge	0.67	2.2	0.38	0.18
GaP	2.25	n/a	0.03	0.015
GaAs	1.42	10^{-6}	0.85	0.04
InSb	0.17	2×10^4	7.7	0.07
CdS	2.40	n/a	0.03	n/a
ZnTe	2.26	n/a	0.03	0.01



Material	*Band Gap (eV)*	*Electrical Conductivity [(Ω-m)⁻¹]*	*Electron Mobility (m²/V-s)*	*Hole Mobility (m²/V-s)*
		Elemental		
Si	1.11	4×10^{-4}	0.14	0.05
Ge	0.67	2.2	0.38	0.18
		III–V Compounds		
GaP	2.25	—	0.03	0.015
GaAs	1.42	10^{-6}	0.85	0.04
InSb	0.17	2×10^4	7.7	0.07
		II–VI Compounds		
CdS	2.40	—	0.03	—
ZnTe	2.26	—	0.03	0.01

Figure 21. Band gap, electrical conductivity and mobilities of Si, Ge, GaP, GaAs, InSb, CdS and ZnTe, copied from Session 13 Slide 33.

Q10 on Slides 34 and 41 asks to order semiconductors by conductivity at the same temperature and to explain the criterion; Slide 34 poses AlP, GaAs and InSb, and Slide 41 the doped pairs (Ge with As or Sb, Si with B or In). It is problem 10 of the sheet.

§ 12 Extrinsic semiconductors: type n and type p [Session 13 Slides 35-39]

12.1 Doping [Session 13 Slide 35]

Extrinsic semiconductors are the most usual type: the dopants add charge carriers, raise the conductivity and make it possible to manufacture devices with special properties (transistors and others). Two types, depending on the doping atoms:

- **Type n:** dopant with **more valence electrons** than the host.
- **Type p:** dopant with **fewer valence electrons** than the host.

12.2 Type n [Session 13 Slide 36]

The impurity atoms have more valence electrons than the atoms of the network, so they behave as **electron donors** (group VA: P, As, Sb; the periodic-table fragment of Slide 35 also lists N). In silicon (4 valence electrons) a phosphorus atom (5 valence electrons) uses four of them in the covalent bonds and has **one excess electron**, loosely bound to it. This electron needs only a small energy to jump into the conduction band. In band language the donor atoms create a **donor level** E_d inside the gap, very close to the conduction band, so the excitation energy is $\Delta E = E_c - E_d$, small compared with E_g. The distance from the donor level to the conduction band determines n_e. The electrons in the conduction band are the **majority carriers**, $n_e \gg n_H$, and

$$\sigma \approx n_e \left|q_e\right| \mu_e$$

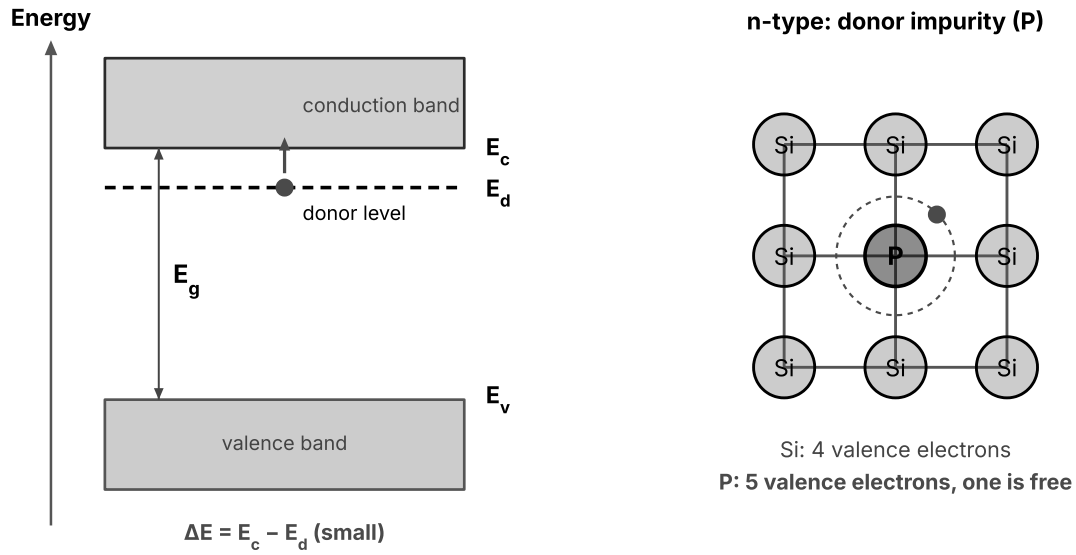


Figure 22. n-type semiconductor: a P atom (5 valence electrons) in Si gives an extra electron in a donor level Ed just below the conduction band, so little energy is needed to free it (schematic, not to scale).

12.3 Type p [Session 13 Slide 37]

The impurity atoms have fewer valence electrons than the atoms of the network, so they behave as **acceptors** (B, Al, Ga and In of group IIIA; in silicon, boron has 3 valence electrons). There is a bond with a missing electron: an **excess of holes**. The acceptor creates an **acceptor level** E_a in the gap, just above the top of the valence band, so a small energy $\Delta E = E_a - E_v$ lifts an electron from the valence band into the acceptor level, leaving a hole in the valence band. The holes moving through the valence band are the majority carriers, $p \gg n$, and

$$\sigma \cong p \left|q_e\right| \mu_H$$

PRINTED GROUP

Slide 37 writes "B, Al, Ga (group VA)", but the table printed on the same slide places B, Al, Ga and In in group 3A (IIIA); group VA is the one of the donors. The acceptors are IIIA.

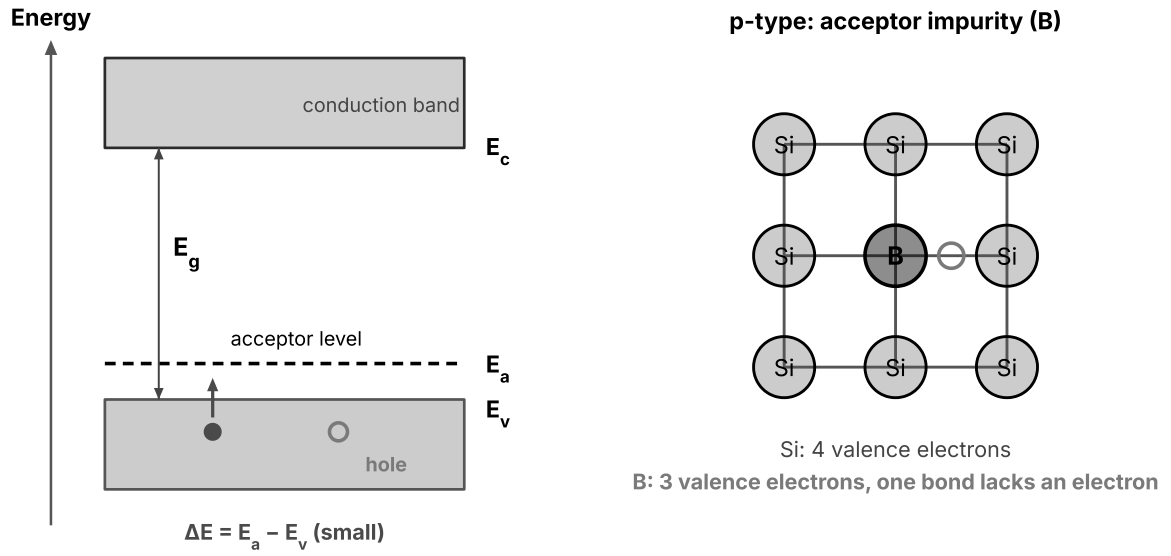


Figure 23. p-type semiconductor: a B atom (3 valence electrons) in Si creates an acceptor level Ea just above the valence band; an electron that jumps into it leaves a hole in the valence band (schematic, not to scale).

12.4 Excitation and majority and minority carriers [Session 13 Slide 38]

In a donor semiconductor the excitation lifts an electron from the donor state to the conduction band, generating a free electron. In an acceptor semiconductor the excitation lifts an electron from the valence band into the acceptor level, generating a hole in the valence band. The slide ends with the table of carrier names:

Semiconductor	Majority-carrier concentration	Minority-carrier concentration
n-type	n_n (electrons in n-type material)	p_n (holes in n-type material)
p-type	p_p (holes in p-type material)	n_p (electrons in p-type material)

12.5 Question 5 [Session 13 Slide 39]

Q5 asks whether silicon doped with antimony, germanium doped with arsenic and silicon doped with gallium (substitutional solid solutions) are n-type or p-type and whether the impurities act as donors or acceptors. The version of the sheet changes the second case to silicon doped with arsenic. It is problem 5 of the sheet, and problem 6 (phosphorus in silicon) applies $\sigma \approx n_e|q_e|\mu_e$.

§ 13 Extrinsic semiconductors: dopants and mobility [Session 13 Slides 40-42]

13.1 Donor and acceptor levels [Session 13 Slides 40-41]

The slide gives the donor and acceptor energy gaps (in electron volts) when silicon and germanium are doped, that is, the energy $E_c - E_d$ or $E_a - E_v$ that an electron needs, and the maximum solid solubility of each dopant:

Dopant	Si: E_d	Si: E_a	Ge: E_d	Ge: E_a
P	0.045		0.0120	
As	0.049		0.0127	
Sb	0.039		0.0096	
B		0.045		0.0104
Al		0.057		0.0102
Ga		0.065		0.0108
In		0.160		0.0112

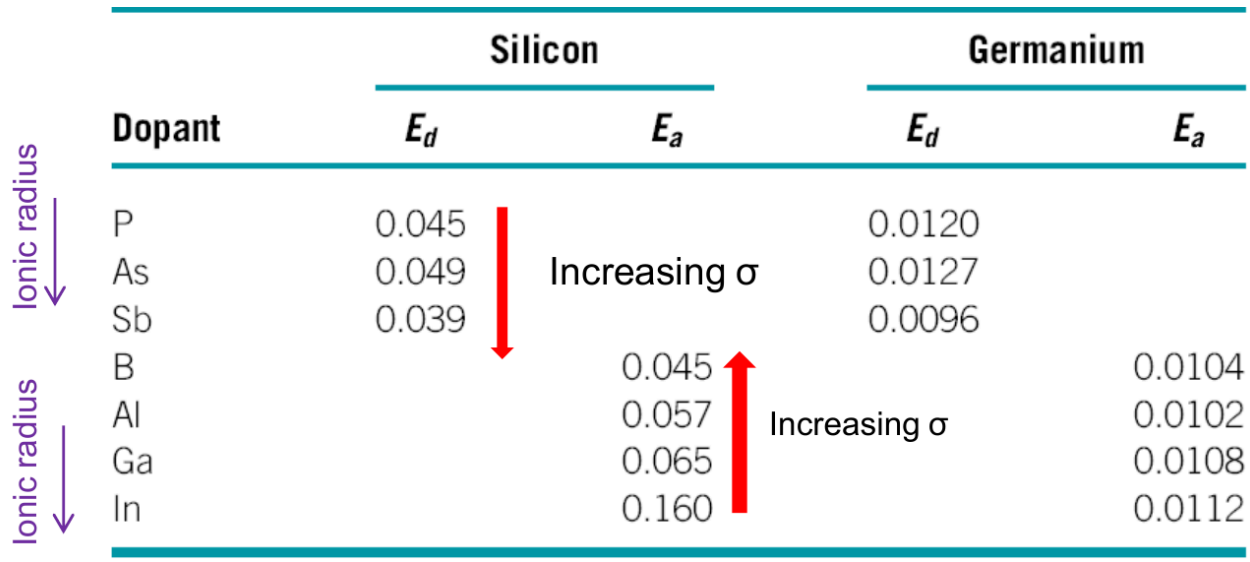
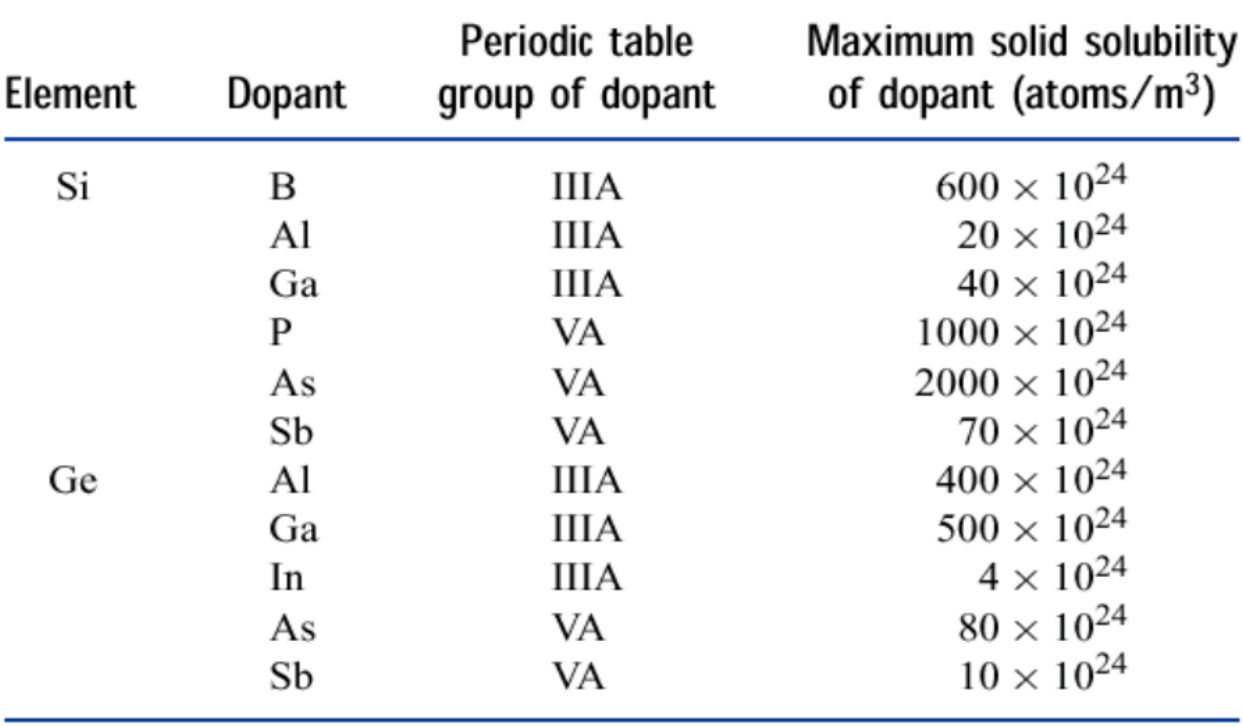


Figure 24. Donor and acceptor energy gaps (in eV) when silicon and germanium are doped, copied from Session 13 Slide 40.

The arrows of the slide say that, as the ionic radius of the dopant increases, the conductivity increases for the donors (P, As, Sb: Si and Ge) and decreases for the acceptors (B, Al, Ga, In), in agreement with the table, where the acceptor energies grow with the size (B $<$ Al $<$ Ga $<$ In). The author's rule of thumb for comparing dopants (not printed on the slide) is to compare E_d or E_a: the smaller the energy, the more carriers at a given temperature. The lower table of the slide gives the maximum solid solubility of the dopants, in atoms/m³:

Host	Dopant	Group	Maximum solubility in atoms/m³
Si	B	IIIA	600×10^{24}
Si	Al	IIIA	20×10^{24}
Si	Ga	IIIA	40×10^{24}
Si	P	VA	1000×10^{24}
Si	As	VA	2000×10^{24}
Si	Sb	VA	70×10^{24}
Ge	Al	IIIA	400×10^{24}
Ge	Ga	IIIA	500×10^{24}
Ge	In	IIIA	4×10^{24}
Ge	As	VA	80×10^{24}
Ge	Sb	VA	10×10^{24}



Element	Dopant	Periodic table group of dopant	Maximum solid solubility of dopant (atoms/m³)
Si	B	IIIA	600×10^{24}
	Al	IIIA	20×10^{24}
	Ga	IIIA	40×10^{24}
	P	VA	1000×10^{24}
	As	VA	2000×10^{24}
	Sb	VA	70×10^{24}
Ge	Al	IIIA	400×10^{24}
	Ga	IIIA	500×10^{24}
	In	IIIA	4×10^{24}
	As	VA	80×10^{24}
	Sb	VA	10×10^{24}

Figure 25. Maximum solid solubility of some dopants in Si and Ge, copied from Session 13 Slide 40.

Slide 41 repeats the same two tables with the second part of Q10: germanium with 5 ppm of As or of Sb, silicon with 10 ppm of B or of In (the slide prints the germanium pair twice).

13.2 Mobility of the carriers [Session 13 Slide 42]

In extrinsic semiconductors also $\mu_e > \mu_H$ (the ratio μ_e/μ_H is 2.81 for Si and 2.05 for Ge). **The distortion of the crystal structure caused by the doping is negligible, so σ is controlled by the number of charge carriers.** The chart shows the room-temperature electron and hole mobilities of silicon against the dopant concentration: both are almost constant (about 0.13 and 0.05 m²/(V·s)) up to about 10^{21} m⁻³ and fall for high doping (electrons about 0.10 m²/(V·s) at 10^{22} m⁻³ and below 0.01 m²/(V·s) at 10^{25} m⁻³; read from the chart).

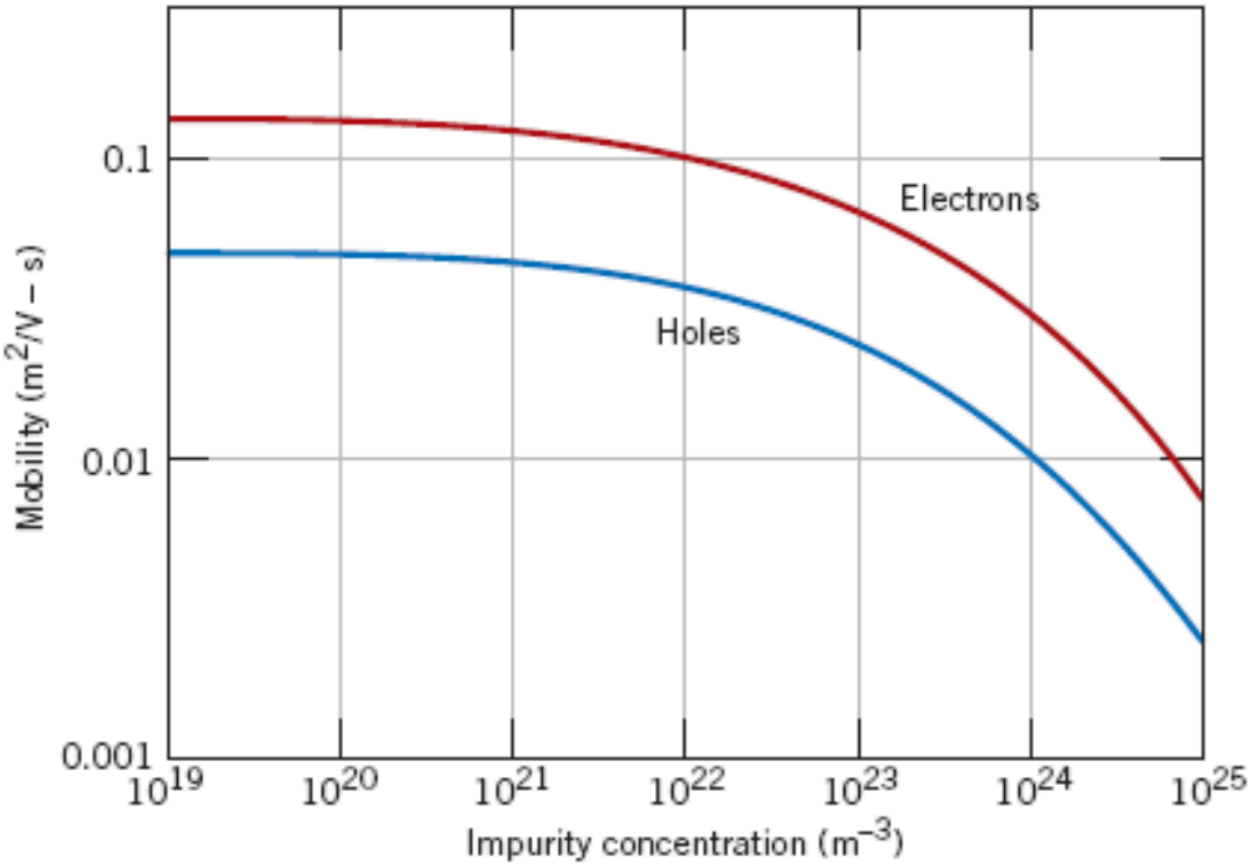


Figure 26. Dependence of the room-temperature electron and hole mobilities on the dopant concentration for Si, copied from Session 13 Slide 42.

§ 14 Extrinsic semiconductors: effect of temperature [Session 13 Slides 43-44]

At **low temperature** the number of ionised impurities per unit volume determines σ. As T increases, the number of ionised atoms increases and so does σ: this is the **extrinsic range**, whose slope in the $\ln\sigma$ against $1/T$ plot depends on the type of semiconductor. For the n-type and p-type the slide writes

$$\sigma = \sigma_0 \exp\left(-\frac{E_c - E_d}{2kT}\right) \qquad \text{and} \qquad \sigma = \sigma_0 \exp\left(-\frac{E_a - E_v}{2kT}\right)$$

so $\ln\sigma = \ln\sigma_0 - \dfrac{E_c - E_d}{2kT}$ for the n-type and the same with $E_a - E_v$ for the p-type: the slope of the line is $-(E_c - E_d)/2k$ (n-type) or $-(E_a - E_v)/2k$ (p-type).

> **PRINTED ENERGY FOR THE P-TYPE**
> Slide 43 prints $E_a - E_c$ for the p-type. The ionisation energy of an acceptor is the distance from the acceptor level to the top of the valence band, $E_a - E_v$ (Slide 37), and that is the form used here.

At **intermediate temperature** all the impurities have been ionised and σ is almost constant: the **exhaustion range** for the n-type (all the donor electrons have been lost) and the **saturation range** for the p-type (all the atoms are ionised). At **high temperature** the intrinsic generation across the gap takes over, with slope $-E_g/2k$: the **intrinsic range** [Session 13 Slide 43]. Slide 44 summarises it: at low and moderate T the material behaves as an extrinsic conductor and σ is governed by the number of ionised atoms; at high T the conductivity slightly decreases, because the number of carriers stays constant and the disordering effect of the temperature (lower mobility) becomes visible; at very high T the material behaves as an intrinsic conductor.

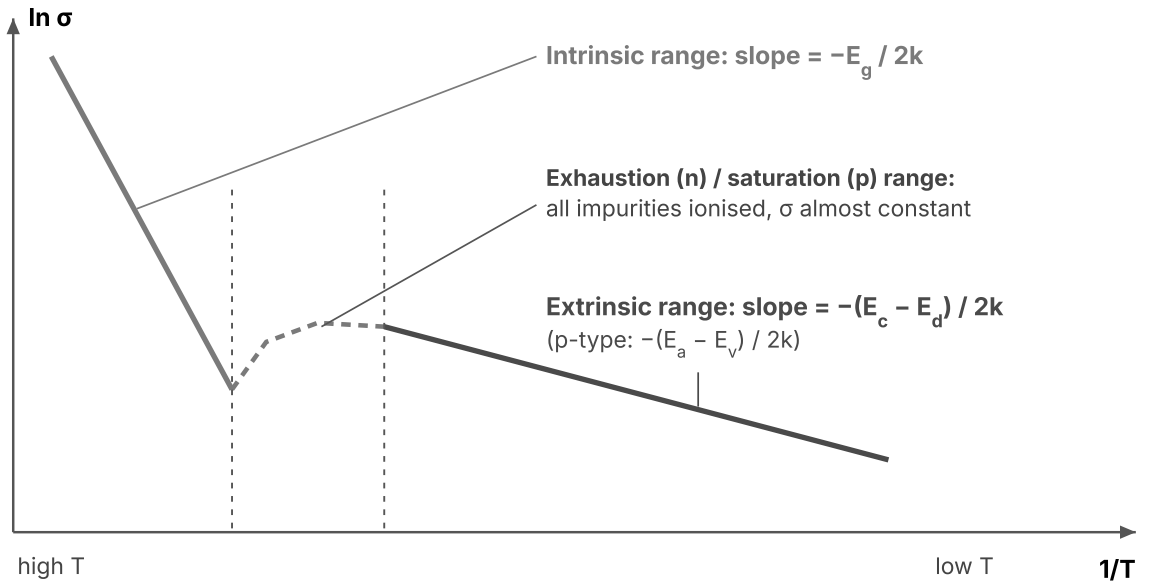


Figure 27. ln σ against 1/T of an extrinsic semiconductor: extrinsic range at low T, exhaustion (n-type) or saturation (p-type) range at intermediate T, intrinsic range at high T (schematic, not to scale).

§ 15 Applications of semiconductors [Session 13 Slides 45-49]

15.1 The p-n junction [Session 13 Slide 45]

A **p-n junction** is a boundary or interface between two types of semiconductor material, p-type and n-type, inside a single crystal. It is the basis of the diode, the photovoltaic cell and the LED of the next sections (the slide links two videos, listed at the end of this page).

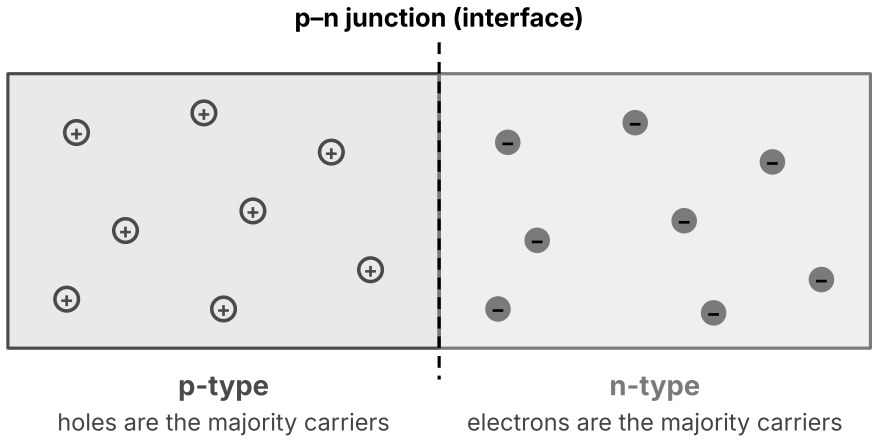


Figure 28. A p–n junction is the boundary between a p-type region (holes) and an n-type region (electrons) (qualitative sketch).

15.2 Photoconductivity and photovoltaic cells [Session 13 Slide 46]

Electrons jump from the valence to the conduction band due to **thermal excitation**, due to **intense electric fields** or due to **exposure to radiation of a given wavelength**. **Photoconductivity** is the phenomenon in which a material becomes more conductive due to the absorption of electromagnetic radiation, such as visible light, whose photon energy must be at least E_g. The same effect used to generate electricity gives the **photovoltaic cells**; solar light has $E_g \sim 1.5$ eV. The slide's absorption figure marks that silicon absorbs photons above 1.1 eV, gallium arsenide above 1.43 eV and aluminium gallium arsenide above 1.7 eV, while photons of lower energy go through.

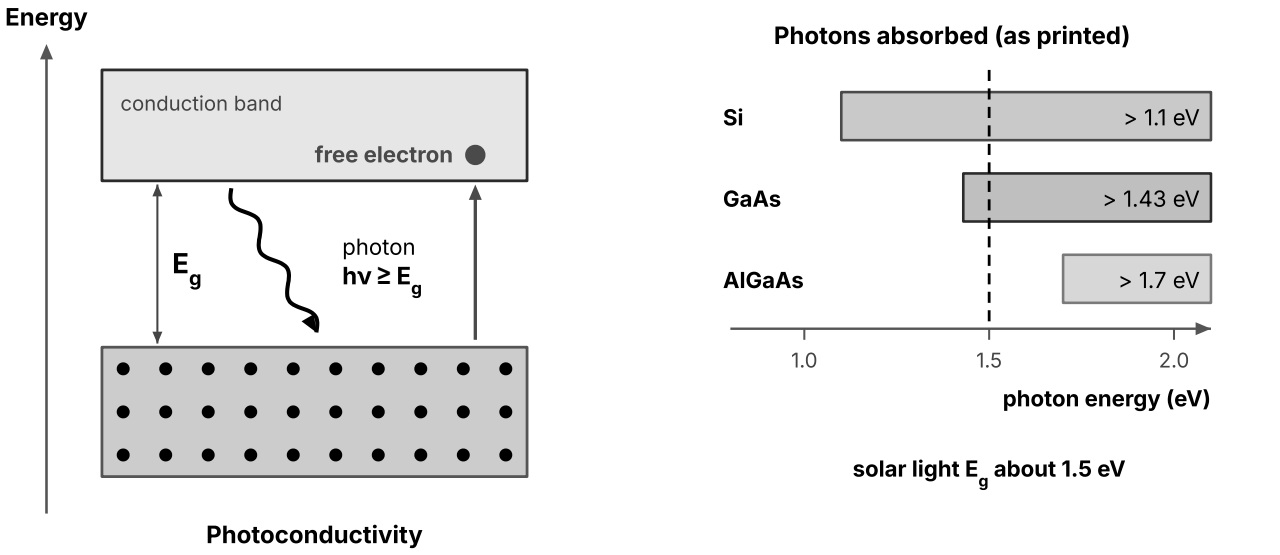
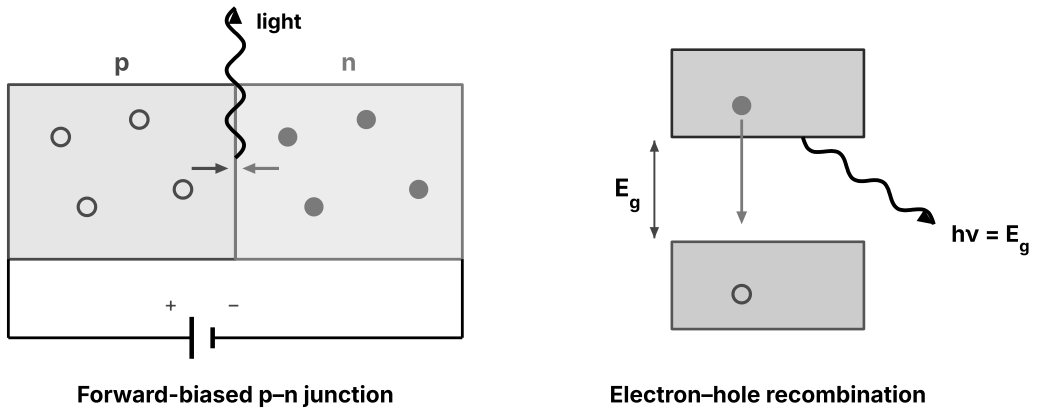


Figure 29. A photon with hv ≥ Eg lifts an electron across the gap, so the conductivity rises; the absorption thresholds of Si, GaAs and AlGaAs and the Eg of solar light (about 1.5 eV) are those printed on the slide (schematic).

15.3 LED [Session 13 Slide 47]

A **light-emitting diode** is a semiconductor doped to create a p-n junction that emits **monochromatic light** when it is directly polarised (forward bias): electrons from the n side and holes from the p side recombine at the junction and release the energy E_g as a photon. The **colour depends on the distance between the valence and the conduction band**, as in the slide's table (the band gap increases from top to bottom and the wavelength increases from bottom to top):

Material	Radiation
AlGaAs	red and infra-red
GaN	green
GaP	green
ZnSe	blue
InGaN	blue
SiC	blue
Diamond	ultra-violet



The colour depends on E_g: red and infra-red (AlGaAs), green (GaN, GaP), blue (ZnSe, InGaN, SiC), ultra-violet (diamond)

Figure 30. LED: in a forward-biased p–n junction, electrons and holes recombine and emit a photon of energy about Eg, so the colour depends on the band gap (schematic).

15.4 Thermistors [Session 13 Slide 48]

Thermistors are devices based on the change of the electrical conductivity of semiconductors with temperature. The chart compares aluminium, whose conductivity decreases with temperature (reducing mobility), with germanium, whose conductivity increases by several orders of magnitude between 0 and 400 °C (increasing charge carriers).

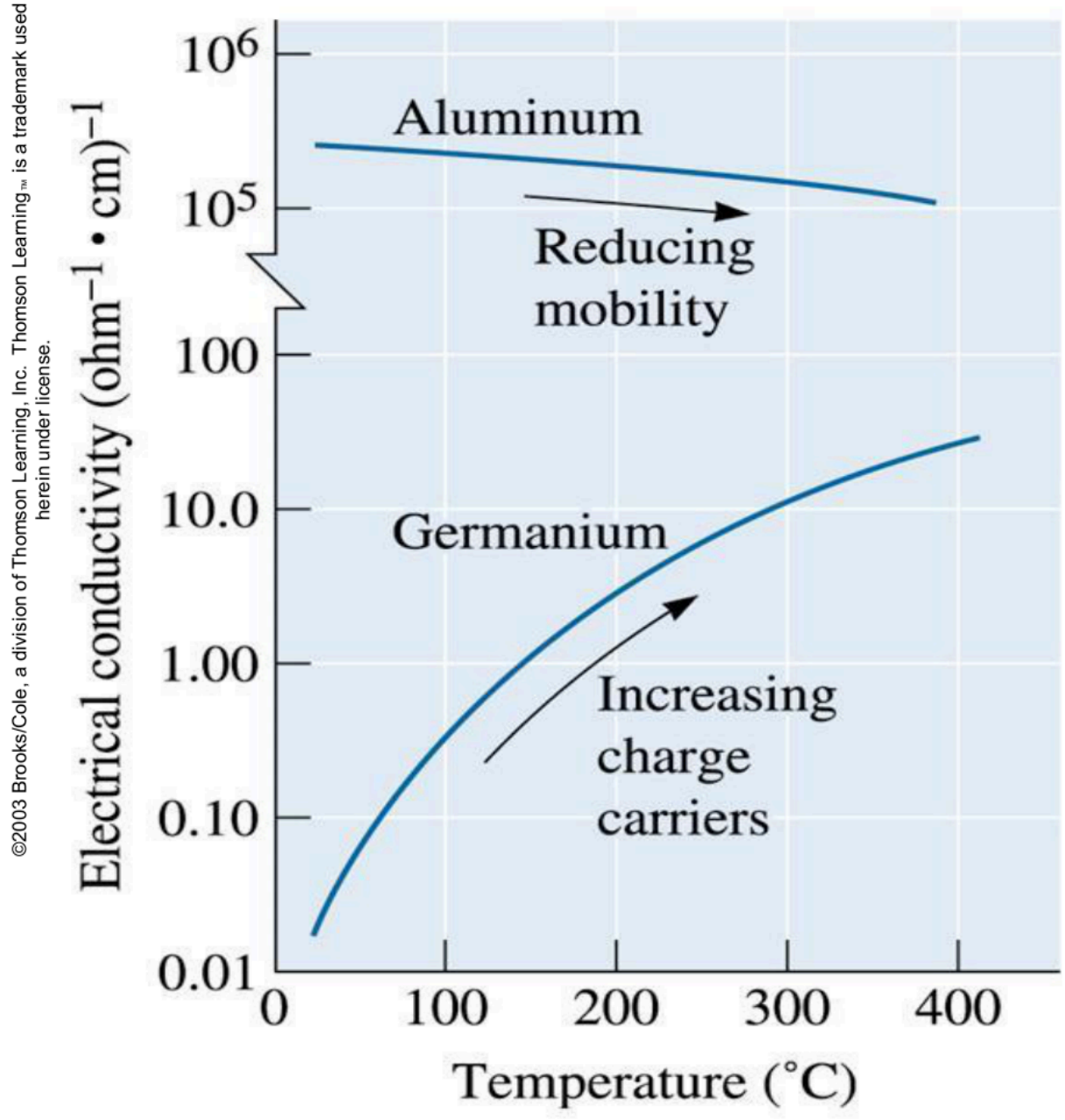


Figure 31. Electrical conductivity against temperature of aluminium (reducing mobility) and germanium (increasing charge carriers), the basis of thermistors, copied from Session 13 Slide 48.

15.5 Transistors [Session 13 Slide 49]

A **transistor** is a semiconductor device that can be used to amplify electrical signals. The slide shows the downscaling of the transistor and of its interconnects through the generations of 180 nm (1999), 130 nm (2001), 90 nm (2003), 65 nm (2005) and 45 nm (2007), down to prototypes of 30 nm and 20 nm, with micrographs and a plot of the size against the year (source: Intel).

§ 16 Insulating and dielectric materials [Session 13 Slides 51–54]

16.1 Dipoles and polarisation [Session 13 Slide 51]

An electrically insulating material (large E_g) that exhibits, or can be made to exhibit, a **dipolar structure** is a dielectric. The **dipole moment** of two charges $+q$ and $-q$ separated by a distance d is

$$p = q \cdot d$$

Without field the dipoles of the material are randomly oriented and the net polarisation is zero (unpolarised); an applied electric field **polarises** the material by aligning the dipoles, and opposite charges appear on the faces next to the plates.

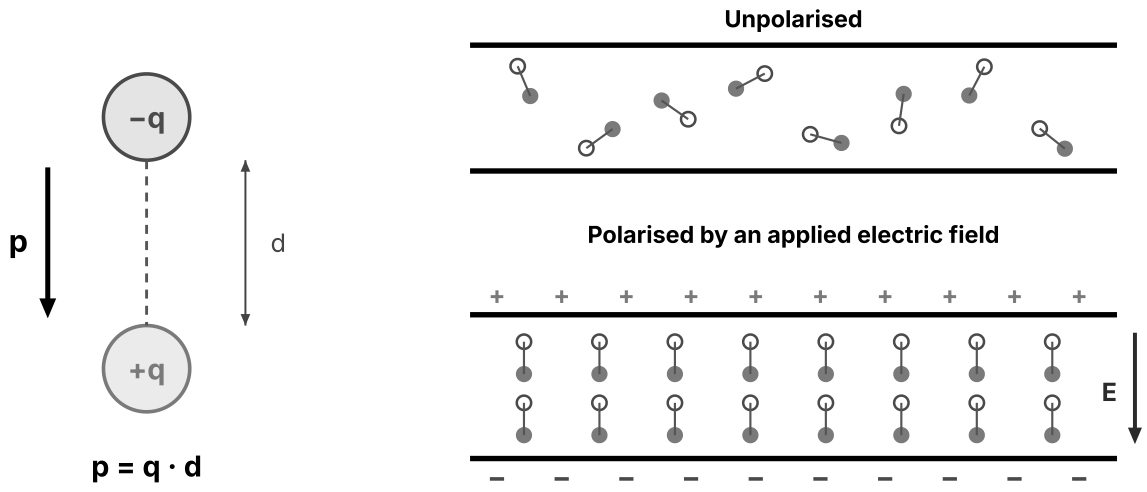


Figure 32. Dipole moment p = q d of two opposite charges a distance d apart; in a dielectric the dipoles are random without a field and line up with an applied electric field (schematic).

16.2 Insulators, dielectrics and capacitance [Session 13 Slide 52]

Highly resistive materials have two applications, depending on their response to the electric field:

- **Insulators** insulate conductors from each other and from a metal mass: high E_g.
- **Dielectric materials** are placed between the plates of a capacitor and increase its capacitance: intense dipoles that easily line up with the electric field.

The capacitance of two parallel plates of area A separated by a distance d is

$$C = \varepsilon_r \, \varepsilon_0 \, \frac{A}{d} \qquad \text{and} \qquad C = \varepsilon_r \, C_0 \quad \text{with} \quad C_0 = \varepsilon_0 \frac{A}{d}$$

$\varepsilon_0 = 8.85 \times 10^{-12}$ F/m is the permittivity of vacuum, ε_r the dielectric constant (relative permittivity) and C_0 the capacitance with vacuum between the plates. The dielectric polarises, so the net charge at its surfaces ($-Q'$ and $+Q'$) lets the plates store more charge, $Q_0 + Q'$, for the same voltage.

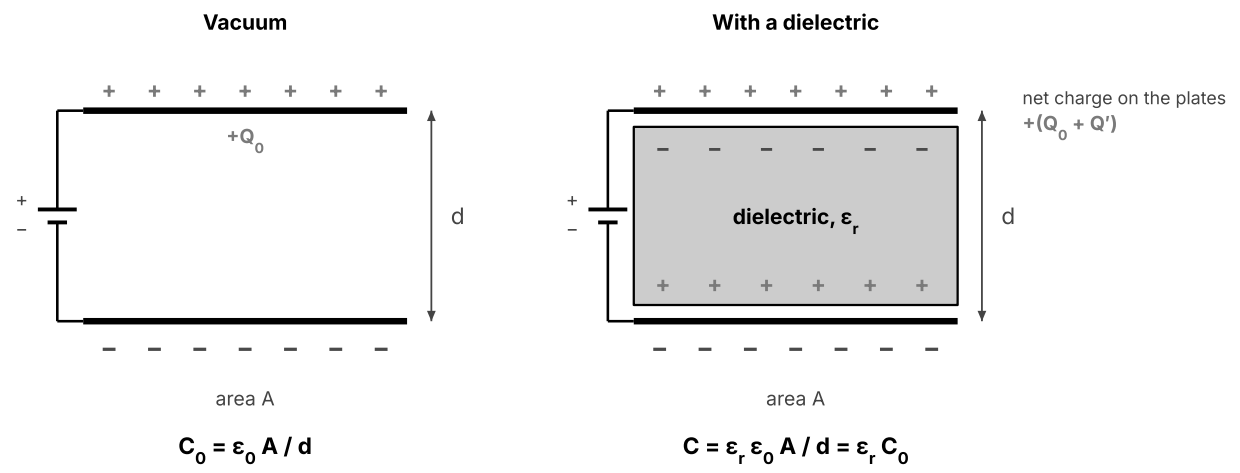


Figure 33. Parallel-plate capacitor in vacuum and with a dielectric: the polarisation charges of the dielectric let the plates hold more charge, so C = εr C0 (schematic).

16.3 Loss factor and dielectric strength [Session 13 Slide 53]

Capacitors are normally used in alternating-current circuits, so the **dipoles must realign fast** when the field changes. The **loss factor** $\tan\delta$ is the amount of electrical energy lost, as heat, by a capacitor in an alternating circuit, due to the friction among dipoles when they try to reorient. It **increases with moisture, temperature and the frequency of the current**, and these also lower the resistivity.

The **dielectric strength** is the maximum electric field that the dielectric can sustain without electrical breakdown:

$$E_{break} = \frac{U_{break}}{d}$$

U_{break} is the breakdown voltage and d the separation between the conductors (in mm on the slide).

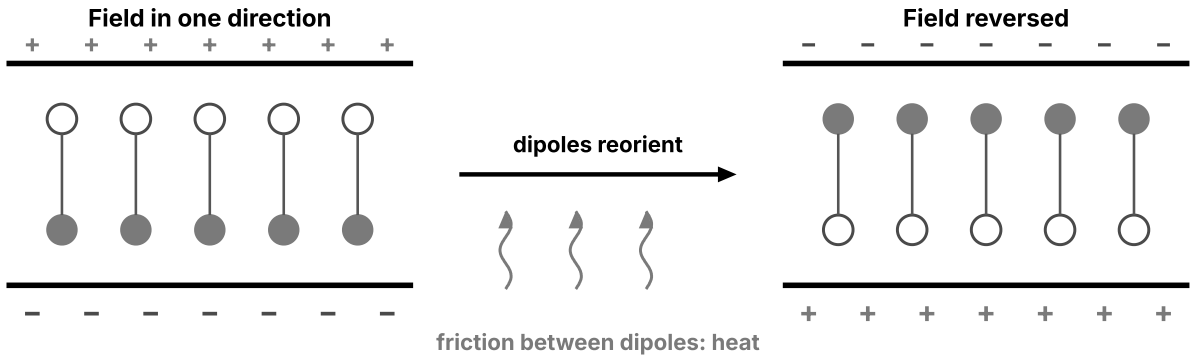


Figure 34. In an alternating current the dipoles of the dielectric must realign at every change of the field; the friction turns electrical energy into heat, measured by the loss factor tan δ (schematic).

16.4 What an insulator should have [Session 13 Slide 54]

Requirement	Why
High resistivity ρ	to avoid current leaks
High dielectric strength E_{break}	to avoid breakdowns of the insulation
Small loss factor $\tan\delta$	(no reason printed; see the loss factor above: energy lost as heat)
Low permittivity ε	to avoid polarisation

The slide closes with a table of typical electrical insulators ("..." marks a value the slide does not give):

Material	Volume resistivity in Ω·m	Dielectric strength in V/mil	Dielectric strength in kV/mm	Dielectric constant K at 60 Hz	K at 10^6 Hz	Loss factor at 60 Hz	Loss factor at 10^6 Hz
Electrical porcelain insulators	10^{11} to 10^{13}	55-300	2-12	6	...	0.06	...
Steatite insulators	$> 10^{12}$	145-280	6-11	6	6	0.008-0.090	0.007-0.025
Fosterite insulators	$> 10^{12}$	250	9.8	...	6	...	0.001-0.002
Alumina insulators	$> 10^{12}$	250	9.8	...	9	...	0.0008-0.009
Soda-lime glass	...	...	...	...	7.2	...	0.009
Fused silica	...	8	...	...	3.8	...	0.00004

§ 17 Ferroelectric materials [Session 13 Slides 55-57]

17.1 Spontaneous polarisation [Session 13 Slide 55]

Ferroelectric materials have **permanent dipoles** that can remain aligned in the absence of an electric field: they show **spontaneous polarisation** with no external field. If a ferroelectric is slowly cooled to a temperature below the critical temperature in the presence of a strong electric field, the anions are displaced towards the positive side of the field and the cations towards the negative side, which produces a permanent electric dipole (the slide draws a perovskite cell with Ti^{4+} or Zr^{4+} at the centre, O^{2-} at the faces and Pb^{2+} at the corners, non-polarised and polarised, for $T < T_{Curie}$).

17.2 Hysteresis and Curie temperature [Session 13 Slide 56]

The electric-field hysteresis cycles are similar to those of ferromagnetic materials with the magnetic field: the polarisation P (or D) against the field E shows a saturation polarisation, a **remanent polarisation** P_r that stays when the field is removed, a non-remanent part, and a coercive field E_c that must be applied in the opposite direction to bring P to zero. The **area under the hysteresis curve** is related to the energy required to change the orientation of the dipoles.

The **Curie temperature** T_c is the temperature above which the material loses its ferroelectric properties and becomes a plain dielectric; typically $-200\,^{\circ}\mathrm{C} < T_c < 640\,^{\circ}\mathrm{C}$. Example: BaTiO3 has $T_c = 120\,^{\circ}\mathrm{C}$. Below 120 °C the structure is **tetragonal**; above 120 °C the crystal structure of BaTiO3 is **cubic** and the permanent dipoles disappear.

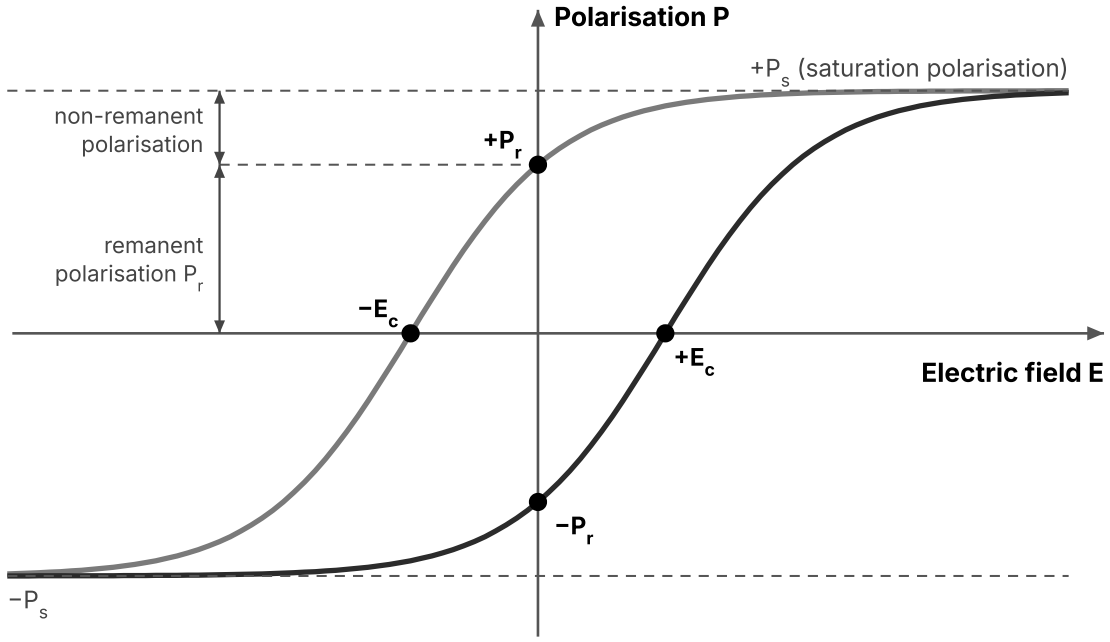


Figure 35. Polarisation against electric field of a ferroelectric: remanent polarisation Pr at E = 0, coercive field Ec, saturation polarisation Ps; the area of the loop is the energy needed to reorient the dipoles (schematic, not to scale).

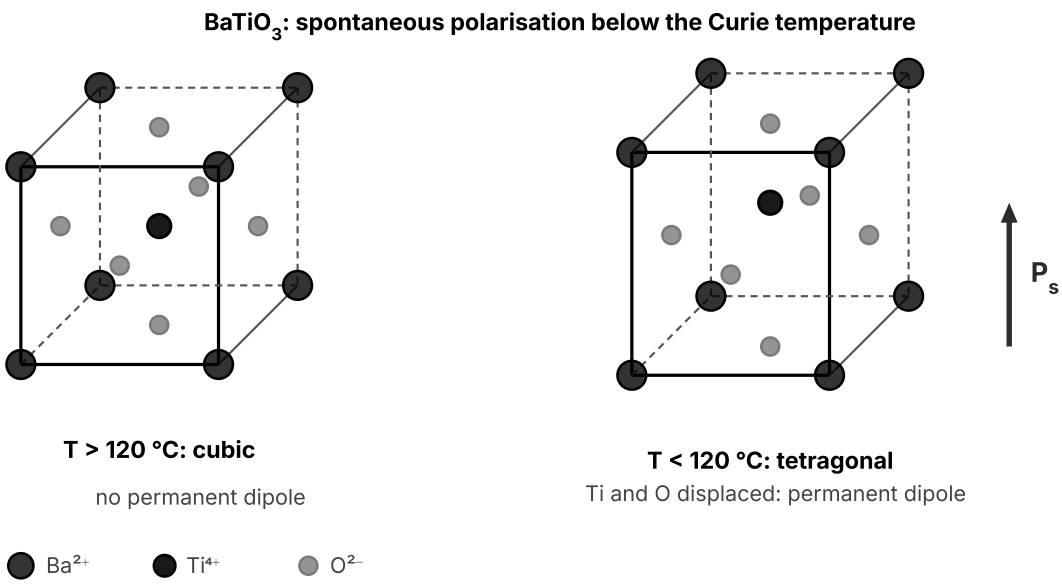


Figure 36. BaTiO3 unit cell: cubic with a centred Ti ion above 120 °C (no permanent dipoles) and tetragonal below it, with Ti and O displaced so that a permanent dipole appears (schematic, displacements exaggerated).

17.3 Dielectric constants [Session 13 Slide 57]

Material	Class	ε_r
$BaTiO_3$	Ferroelectric	4000
$KNbO_3$	Ferroelectric	700
Rochelle salt	Ferroelectric	170
TiO_2	Dielectric	100 to 50
Al_2O_3	Dielectric	10
Porcelain	Dielectric	6.0
PET	Dielectric	3.2
PVC	Dielectric	3.5
Polyethylene	Dielectric	2.3
Air	Dielectric	1.0

$\varepsilon_r(\text{ferroelectric}) \gg \varepsilon_r(\text{common dielectric})$, so ferroelectrics give **small and very efficient capacitors** (for example, for computer memories). Since $C = \varepsilon_r C_0$, a capacitor with BaTiO3 stores about 4000 times the charge of the same capacitor with vacuum.

§ 18 Piezoelectric materials [Session 13 Slides 58-60]

Piezoelectric materials are ferroelectric materials that can be polarised when an external mechanical stress is applied. They occur in ceramics whose crystal structure has a **low degree of symmetry** [Session 13 Slide 58]. Two effects:

Direct piezoelectric effect	Reverse piezoelectric effect (electrostriction)
External stress	External electric field
↓ crystalline structure deformation	↓ dipole alteration
↓ dipole alteration	↓ crystalline structure deformation
Electric field (voltage) generation	Dimensional change of the material

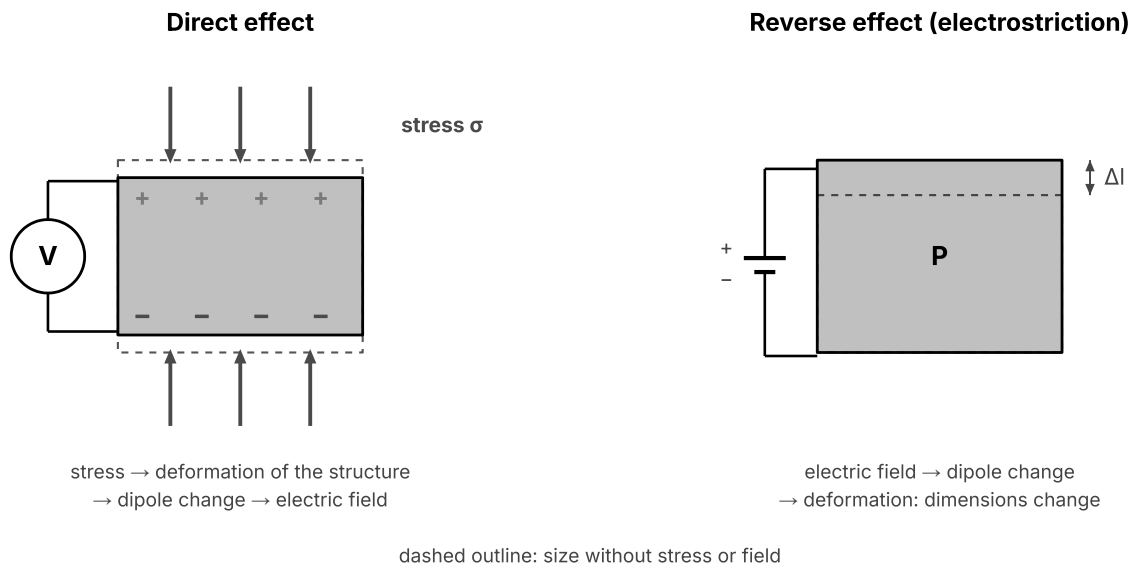


Figure 37. Piezoelectric material: an external stress changes the dipoles and generates an electric field (direct effect), and an external electric field deforms the structure (reverse effect) (schematic).

They are used to transform electrical energy into mechanical energy and vice versa [Session 13 Slide 59], for example in **lighters** (a mechanical shock converted into a spark), **transducers in speakers**, **inkjet printers** (a piezoelectric actuator pushes the ink droplet out of the nozzle with a pulsed voltage), and **microphones and hydrophones** [Session 13 Slide 60].

§ 19 Collected formulas of the topic

Everything below is taken from the sections above; it is a quick reference, not new content.

Quantity	Formula	Where
Resistance of a bar	$R = \rho l/A$ (derived from $J = \sigma E$)	Section 4
Drift velocity and mobility	$v_d = \mu E,\ \mu = \tau \lvert q_e \rvert / m_e$	Section 4
Conductivity of a conductor	$\sigma = n \lvert q_e \rvert \mu = 1/\rho$	Section 4

Quantity	Formula	Where
Matthiessen's rule	$\rho_{total} = \rho_T + \rho_r = \rho_T + \rho_i + \rho_d$	Sections 5, 6
Thermal resistivity of a metal	$\rho_T = \rho_0(1 + \alpha_T \Delta T)$, ρ_0 at 0 °C	Section 5
Intrinsic semiconductor	$\sigma = n\lvert q_e\rvert(\mu_e + \mu_H)$, $\sigma = \sigma_0 e^{-E_g/2kT}$	Section 10
Extrinsic n-type, p-type	$\sigma \approx n_e\lvert q_e\rvert\mu_e$, $\sigma \cong p\lvert q_e\rvert\mu_H$	Section 12
Extrinsic range	$\sigma = \sigma_0 e^{-(E_c - E_d)/2kT}$ or $e^{-(E_a - E_v)/2kT}$	Section 14
Dipole moment	$p = q\,d$	Section 16
Capacitance with dielectric	$C = \varepsilon_r \varepsilon_0 A/d = \varepsilon_r C_0$	Section 16
Dielectric strength	$E_{break} = U_{break}/d$	Section 16

Videos linked in the slides

These are references printed on the slides, not content of the portal:

- Session 13, Slide 45 (p-n junction): https://www.youtube.com/watch?v=btOIDQeMrMg
- Session 13, Slide 45 (p-n junction): https://www.youtube.com/watch?v=ar7xDMR4P_U

See also the solved problems of this topic: Problem - T6-01 Silver and Aluminium Wires with the Same Resistivity, Problem - T6-02 Strategies to Strengthen an Aluminium Wire and Their Effect on its Conductivity, Problem - T6-03 Ionic Ceramic against Metal Defects Temperature and Doping, Problem - T6-04 Effect of Temperature on a Metal and on an Intrinsic Semiconductor, Problem - T6-05 Extrinsic Type of Silicon Doped with Sb As and Ga, Problem - T6-06 Phosphorus-Doped Silicon Type and Conductivity, Problem - T6-07 Conductivity of Intrinsic Germanium at 75 C, Problem - T6-08 Carrier Concentrations of Intrinsic Silicon from its Conductivity, Problem - T6-09 Ranking Semiconductors by Conductivity Temperature Doping and Concentration, Problem - T6-10 Ranking Semiconductors by Conductivity Band Gap and Donor and Acceptor Levels.